

Kobon Triangle Constructions: Uniform Lower Bounds, Extension Methods, and Formal Verification

Alejandro Zarzuelo Urdiales

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Abstract

The Kobon triangle problem asks for the largest number of bounded triangular cells determined by a prescribed number of straight lines in the real affine plane. We give a unified account of a construction and verification program begun in March 2026, distinguishing completed geometric theorems, exact finite certificates, conditional families, and unresolved extension claims. The principal completed all-order result is a simple-line construction with at least $\lfloor n(n-3)/3 \rfloor + 1 + (n \bmod 2)$ triangles for every $n \geq 4$, together with the elementary smaller cases. A phase refinement and changes of affine chart supply a constant-term improvement of the familiar Füredi–Palásti lower formula. We prove its geometry and counting in Lean and combine it with 138 exact finite certificates, retaining the authorship of published inputs. Further results include a verified exterior-line construction, an obstruction to unrestricted repetition of full exterior gains, a geometric one-step specialization of Bartholdi–Blanc–Loisel doubling, and local multiplicity-sensitive lemmas for upper-bound arguments. We document the earlier values at 28, 30 and 34 lines, the finite and proposed infinite families, unsuccessful searches, and the precise remaining proof obligations. The affine numerical refinement is compared carefully with the original projective counts and stronger published subsequences; numerical first priority and an unrestricted successor theorem are not asserted. All figures are generated anew from recorded formulas, coordinates, or experiments.

Keywords. Line arrangements; triangular cells; Kobon problem; constructive lower bounds; projective transformations; exact computation; Lean formalization.

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1 Introduction and scope of the results

An arrangement of n straight lines divides the real plane into bounded and unbounded cells. The Kobon problem asks how many of the bounded cells can be triangles. The difficulty is not creating many triples of intersecting lines: there are $\binom{n}{3}$ candidate triples. The difficulty is ensuring that their triangles are not cut by any of the remaining lines and that their interiors do not overlap. The distinction between an attractive picture and a valid arrangement is therefore central to both construction and verification.

Let $K(n)$ denote the classical maximum, allowing multiple intersections and parallel lines, and let $K_s(n)$ denote the maximum over simple arrangements: no parallel pair and no three concurrent lines. All lower-bound witnesses constructed by the universal theorem below are simple, so they establish lower bounds for both quantities. Some of the strongest retained finite certificates are nonsimple. We never use an upper bound restricted to K_s as an upper bound for K without an additional argument.

This paper consolidates the author’s March 2026 extension proposal and the subsequent program of exact construction, comparison with the literature, and formal proof. The purpose is both mathematical and documentary: to state the strongest completed results, to make their proofs and provenance inspectable, and to preserve the failed approaches that determine the next research questions. The original general extension claim is reported as an unresolved claim rather than silently promoted to a theorem.

1.1 The main completed all-order theorem

Define $B(n) = \lceil n(n-3)/3 \rceil$ for $n \geq 3$, and set $B(n) = 0$ for $n < 3$. The familiar Füredi–Palásti construction supplies this affine baseline; the underlying construction and its historical priority belong to the earlier literature [6, 10]. Our principal completed theorem is the following refinement.

Theorem 1.1 (Universal affine refinement). *For every $n \in \mathbb{N}$, there is a simple arrangement of n real straight lines with at least $G(n)$ bounded triangular cells, where*

$$G(n) = \begin{cases} 0, & 0 \leq n < 3, \\ 1, & n = 3, \\ \lfloor n(n-3)/3 \rfloor + 1 + (n \bmod 2), & n \geq 4. \end{cases} \quad (1)$$

Consequently $K(n) \geq K_s(n) \geq G(n)$.

The proof, developed in Section 4, combines a shifted trigonometric arrangement with one- and two-cap changes of affine chart. The Lean declaration `Kobon.Universal.baseline_sound` proves the geometric existence statement for arbitrary n . It has no assumed extension theorem and no dependence on finite experimental verification. Its dependencies are the standard logical axioms used by Lean and Mathlib.

For $n \geq 4$, the gains over $B(n)$ are 1, 1, 0, 2, 0, 1 in residue classes 0, 1, 2, 3, 4, 5 modulo six. Thus the refinement is strict at every odd order at least five and every positive multiple of six. Both formulas have leading behavior $n^2/3 - n + O(1)$: this is a constant-term refinement, not an improved quadratic or linear asymptotic coefficient. The original Füredi–Palásti projective count table already contains related constants. The present source audit does not establish first numerical priority for the affine formula.

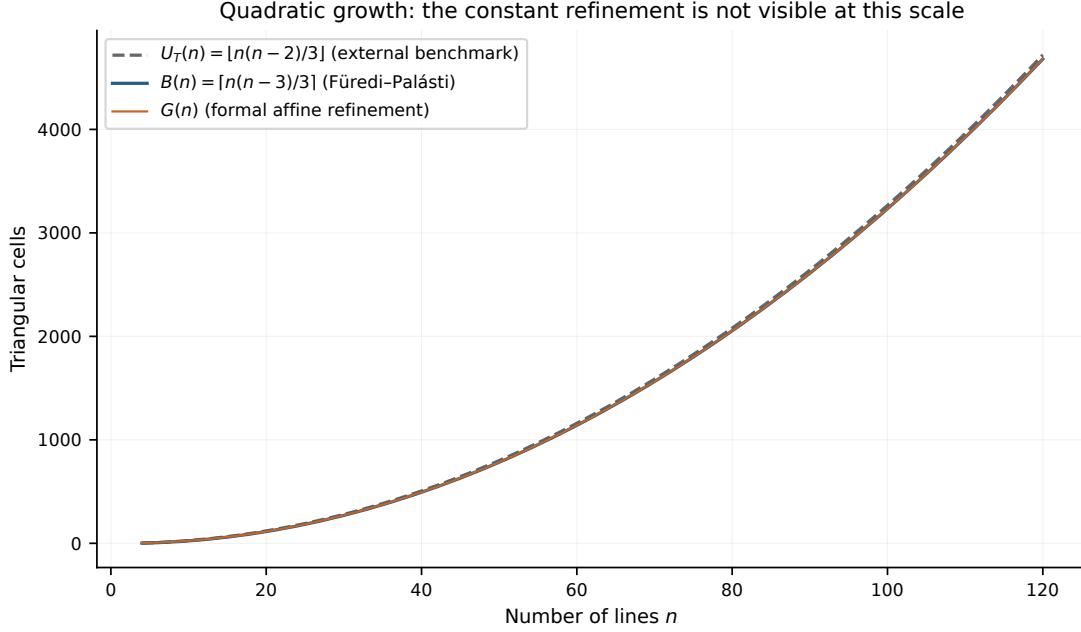


Figure 1: Growth of the classical general baseline, the proved affine refinement, and the classical polynomial upper benchmark. The close overlap on a quadratic scale is mathematically meaningful: the new uniform gain is bounded. Residue and deficit plots below resolve the smaller differences. This figure is computed from the formulas, not copied from earlier publications.

1.2 Finite enhancements and other completed components

The universal construction is supplemented by exact coordinates. Let $C(n)$ be the largest triangle count among the finite certificates in the release at order n , with $C(n) = 0$ when no such certificate is saved. Then

$$L(n) = \max\{G(n), C(n)\}, \quad K(n) \geq L(n) \quad (2)$$

is proved for all natural orders. This elementary maximum operation retains published constructions and the author’s earlier values without altering their attribution. It is not a claim that L is the largest lower bound obtainable from all published constructions by every possible interpolation or extension.

The finite catalog contains 138 distinct coordinate certificates, including 104 simple witnesses. A separate generated table has 64 orders at which a saved finite certificate strictly improves the older baseline B ; some of these improvements are already supplied by G . The catalog includes weaker checkpoints and independent reproductions. Its size is not a count of original discoveries.

The other completed components are an actual exterior-line construction from certified visible pairs; a finite boundary-resource obstruction to repeatedly obtaining the full proposed successor gain; a geometric BBL doubling step under specified seed hypotheses; and local geometric lemmas about fans at multiple intersections. Their scope is summarized in Table 1. The distinction between a proved local step and a recursively closed infinite family is maintained throughout.

1.3 History, attribution, and the structure of the paper

The March archive records a proposed extension from odd to even orders, associated finite bounds, and an incomplete Lean file. That file contains eight proof placeholders, including its

Result	Mathematical content	Release status
Universal G Envelope L	Simple real-line witnesses at every order Maximum with all saved finite witnesses	General Lean proof Lean; finite native checks
28:238, 30:275, 34:357	Earlier attributed finite inequalities	Exact Lean certificates
Exterior extension Full successor recurrence 49-seed numerical target	New line and gain from a visible-pair list $K(n+1) \geq K(n) + \lfloor n/2 \rfloor$ Lower function obtained by summing the proposed recurrence	General Lean proof Unproved Independently proved
One-step BBL doubling	$(q+1, T) \mapsto (2q+1, T+q^2)$ for compatible seeds	General Lean proof
Infinite hybrid iteration	Preservation and repeated use of the seed invariant	Incomplete Lean integration
Local fan structure	Incidence, antipodal and run-length restrictions	General Lean proofs
Global multiplicity consequences	Upper-budget inequalities from geometric charging	Paper-level; bridge open

Table 1: Claim status in the frozen mathematical release. “General Lean proof” does not imply that every proposed application has been formalized. Published construction inputs retain their original authorship.

central extension step. Later exact certificates establish the finite inequalities independently, while the current all-order proof uses a different construction. Accordingly, a corrected account must separate the survival of numerical conclusions from the validity of the original proof strategy.

The September work broadened the program to both parities, exterior visibility, boundary-defect estimates, compatible doubling seeds, exact rational realization, and multiplicity-sensitive upper arguments. Several plausible routes failed: direct repetition of exterior gains lost the necessary boundary resources; small perturbations of nonsimple records sometimes reduced the triangle count; and large local searches failed to improve their chosen seeds. These negative findings are recorded with their exact scope rather than converted into impossibility theorems.

Section 3 states the geometric definitions and the certificate-to-cell bridge. Section 4 proves the uniform refinement; Section 5 records finite cases and provenance. The following sections develop extension and family results, local upper-bound ingredients, and computational searches. The final sections describe the formal trust boundary and future obligations. The appendices contain the complete finite tables and a map from mathematical claims to proof sources. The code and evidence are available at <https://github.com/alejandrozu/kobon-proof> [29].

2 Literature, provenance, and the scope of comparison

2.1 Four distinctions that affect every comparison

The literature studies several related extremal functions. Their numerical tables cannot be combined without retaining their hypotheses. Throughout this paper, $K(n)$ concerns straight lines in the real affine plane and permits multiple intersections and parallel pairs, whereas $K_s(n)$ restricts to pairwise nonparallel lines with no three concurrent. Only bounded triangular cells are counted. A pseudoline arrangement is a topological relaxation; a lower bound

obtained there becomes a straight-line lower bound only after realizability has been established. Finally, projective triangular faces need not remain bounded after an affine chart is chosen.

Distinction	What transfers immediately	What requires another argument
Simple / unrestricted	A simple example is an unrestricted example.	A simple upper bound need not bound a configuration with concurrent lines.
Straight / pseudoline	Every straight arrangement is a pseudoline arrangement.	A pseudoline construction requires a straight realization.
Affine / projective	Projective transformations preserve incidence and projective faces.	Boundedness depends on the line chosen as infinity.
Finite / infinite	A certified configuration proves its individual lower bound.	Iteration requires the next configuration to satisfy every input invariant.

Table 2: Logical distinctions used throughout the source audit.

There are also two different projective-to-affine operations. Choosing a new line at infinity outside an n -line arrangement keeps n lines, but may lose triangular faces to infinity. Making one of the arrangement’s own lines the line at infinity and removing it produces an $(n - 1)$ -line affine arrangement. The latter operation underlies several optimal subsequences; the former is central to the uniform construction formalized here.

2.2 From the puzzle to arrangement theory

Fujimura’s puzzle reached English readers through *The Tokyo Puzzles*, edited by Gardner and translated by Adachi [9]; Gardner also discussed it in his collected mathematical recreations [11]. The broader theory of line and pseudoline arrangements is developed in Grünbaum’s monograph [12]. The papers of Strommer and Purdy document an established extremal-triangle literature preceding the construction used in this paper [22, 23, 26]. We cite those early papers as historical context rather than silently identifying all their projective or non-simple questions with the affine function K .

The familiar comparison polynomial is

$$U_T(n) = \left\lfloor \frac{n(n-2)}{3} \right\rfloor. \quad (3)$$

The attribution to Saburo Tamura is transmitted in the Clément–Bader draft [3]; an original publication by Tamura was not located in this audit. We therefore do not manufacture an independent bibliographic item for that attribution. Clément and Bader report the stronger classical envelope $U_T(n) - 1$ for $n \equiv 0, 2 \pmod{6}$. Section 8 explains both the scope of this reported result and a local proof-reading issue. A source-reported upper bound and an upper bound proved in our Lean development are different kinds of evidence.

2.3 The Füredi–Palásti all-order construction

Füredi and Palásti gave highly triangular straight-line arrangements using trigonometric coordinates [10]. Their projective Table 1 has a residue-dependent constant term. The familiar affine consequence is commonly written

$$B(n) = \left\lceil \frac{n(n-3)}{3} \right\rceil \quad (n \geq 3). \quad (4)$$

Erickson states this formula explicitly in Section 7.2 while using arrangements in a computational-geometric lower-bound argument [6]. Felsner and Kriegel place the construction in the study of Euclidean triangular cells [7]. A later discussion by Damásdi, Martínez-Sandoval, Nagy and Nagy quotes the slightly weaker floor version while studying triangle areas [4]. Thus a general lower bound valid at every order existed well before the present work.

Our completed formula is

$$G(n) = \left\lfloor \frac{n(n-3)}{3} \right\rfloor + 1 + (n \bmod 2), \quad n \geq 4. \quad (5)$$

The constructions in later sections prove this for actual simple affine lines. Table 3 separates its arithmetic improvement over (4) from the projective count in the original paper. Here $P_{\text{FP}}(n)$ denotes the latter construction’s projective count, not the projective maximum.

$n \bmod 6$	$G - B$	$P_{\text{FP}}(n), n \geq 5$	$P_{\text{FP}} - G$
0	1	$n(n-3)/3$	-1
1	1	$(n^2 - 3n + 5)/3$	0
2	0	$(n^2 - 3n + 8)/3$	2
3	2	$(n^2 - 3n + 9)/3$	1
4	0	$(n^2 - 3n + 8)/3$	2
5	1	$(n^2 - 3n + 5)/3$	0

Table 3: Exact comparison with the quoted affine baseline and with the original projective construction’s face counts. A projective surplus does not itself supply additional bounded affine triangles.

The numerical gain is bounded by two. In particular, the quadratic coefficient $1/3$ and linear coefficient -1 remain unchanged. What the completed proof adds is a phase refinement, a controlled choice of affine chart for odd orders, and an end-to-end geometric formalization. The audited sources do not provide an explicit statement of (5) with this affine proof. That observation is not an exhaustive numerical-priority theorem: old projective constructions contain related counts, and an unnoticed affine-chart argument could have been available earlier. The defensible claim is the proved improvement over the explicitly quoted formula (4), together with the formal construction; a claim of first numerical discovery is withheld.

2.4 Optimal subsequences and the straightening problem

Harborth and Roudneff developed constructions and extremal results for pseudolines [14, 24]. Their scope matters because topological realizability does not imply straight realizability. Forge and Ramírez Alfonsín constructed a projective straight-line family at orders $2 \cdot 2^t + 2$ [8]. Removing a distinguished line at infinity gives optimal affine orders $2 \cdot 2^t + 1$; in particular, this is the relevant older source for the 33-line order.

Bartholdi, Blanc and Loisel’s preprint appeared in 2007 and its proceedings publication in 2008 [1]. Their compatible doubling step takes $q + 1$ lines to $2q + 1$ and adds q^2 triangles. It requires a prescribed tangent grid and a fully used distinguished line. Their straight-line optimal series include $14 \cdot 2^t + 1$ and $6 \cdot 2^t + 1$; their $18 \cdot 2^t + 1$ series was then a pseudoline result. The general numerical recurrence is therefore prior work, not a new theorem of this project. Our contribution to that step is the verified real-coordinate specialization and its explicitly recorded hypotheses.

Parpalak and Utkin supplied a compatible straight 19-line seed and established the straight-line series $n = 18 \cdot 2^t + 1$ [18]. Its triangle count is $108 \cdot 4^t - 1$. Substitution into our

formula gives the exact comparison

$$(108 \cdot 4^t - 1) - G(18 \cdot 2^t + 1) = 6 \cdot 2^t - 2. \quad (6)$$

Thus their series is stronger at these orders: the comparisons begin $107 > 103$ at 19, $431 > 421$ at 37, and $1727 > 1705$ at 73. Their separate even-series draft is also relevant prior construction evidence [19]. Its numerical families are not replaced by a weaker universal formula.

It would consequently be misleading to call G the strongest numerical lower bound for every n , or an unqualified state of the art among all possible all-order formulas. Taking a maximum with known special constructions already improves it. Padding a configuration by lines sufficiently far away preserves its bounded triangles and extends each special construction to later orders. The retained function in the repository likewise takes a maximum with its finite certificate catalogue. The purpose of G is a compact uniform guarantee with a complete formal proof, not the suppression of stronger special cases.

2.5 Upper-bound refinements retain their simplicity assumptions

The even-order bound

$$\left\lfloor \frac{n(n - 7/3)}{3} \right\rfloor$$

in Bartholdi–Blanc–Loisel applies to *simple affine pseudoline* arrangements. Blanc subsequently proved the sharper common even-order polynomial

$$K_s(n) \leq \left\lfloor \frac{n(n - 5/2)}{3} \right\rfloor, \quad n \text{ even}, \quad (7)$$

also in the simple pseudoline setting [2]. The latter work was posted in 2008 and published in 2011. Its parity argument is the starting point for our clean-line accounting outside a multiple-point core.

The distinction is concrete. Formula (7) equals 53 at $n = 14$, whereas a nonsimple configuration with 54 triangles exists. Applying a simple upper theorem to that configuration would be invalid. Our upper-bound work therefore counts shared sides and multiple incidences explicitly rather than treating a small perturbation to general position as automatically harmless.

2.6 Recent computational constructions and versioned claims

Savchuk’s 2025 work combines arrangement tables, SAT search, and heuristic straightening [25]. It reports optimal 23- and 27-line configurations and an exclusion of a perfect 33-triangle arrangement at order 11. Its Appendix A.4 records the earlier Honma-based 11-line construction. For our purposes, an arrangement table, a floating-point fit, an exact straight-line certificate, and an exhaustive upper-bound exclusion are four different outputs. The present work independently verifies the coordinate lower bounds it imports; it does not thereby replay the external SAT exclusions.

Maiorana released fifteen exact 14-line, 54-triangle configurations [16]. We preserve their attribution and the source’s CC BY 4.0 license. The current Parpalak–Utkin gallery supplies further exact constructions, including nonsimple even-order examples [20]. The 8-line construction has earlier provenance; citing its current host does not transfer its invention to the host authors. Each retained dataset is tied to a source revision, checked independently, and distinguished from any stronger optimality language used in its original repository.

Versioning also matters for exclusions. The fourth version of Liang, Liu and Zhang’s preprint concerns symmetry restrictions for simple 28-line configurations [15]. It is not treated here as a proof of an unrestricted exact value, and its displayed numerical comparison is not substituted for Blanc’s simple upper theorem. The ProofAtlas research page similarly reports restricted upper results while keeping the unrestricted problem open [21]. We record these as current research sources, not as independently reproduced proofs.

2.7 The March result, later verification, and attribution

The author’s March manuscript proposed an even-order extension and recorded the values 238, 275, and 357 at orders 28, 30, and 34[28]. These values are now supported in this project by exact simple-line certificates. The associated original Lean file, however, contained placeholders, including the general extension lemma. The later successful formalizations do not retroactively turn that unfinished universal argument into a proof.

OEIS A006066 indexes numerical data, references, and named contributions [17]. It is not a complete priority database. In particular, the 2007 Bartholdi–Blanc–Loisel table already contains a bold, hence straight-realizable, 30:275 entry. The 34:357 value follows from the older perfect 33-line family and its exterior extension. At 28 lines, Savchuk describes the analogous extension of the 27-line solution. These overlaps require a distinction between an independently supplied construction, a new exact or formal certificate, and the first known numerical lower bound. We retain the author’s recorded contribution without claiming first numerical priority solely from the OEIS attribution.

The recurrence reproduced by HandWiki [13] is likewise a record of the proposed claim rather than a substitute for its proof. The unrestricted statement

$$K(n+1) \geq K(n) + \lfloor n/2 \rfloor$$

has not been established by the work reported here. What has been established includes a geometric exterior extension under explicit visibility hypotheses, finite successful successor steps, and a separate unconditional all-order construction. Their logical roles remain distinct.

2.8 What formal verification contributes

The proof infrastructure uses Lean 4 [5] and mathlib [27]. For the universal result, it connects analytic line equations, determinant signs, triangular cells, affine-chart control, and finite counting, rather than checking the final arithmetic formula in isolation. For finite results, it turns attributed coordinates into explicit certificates. For the upper-bound investigation, it proves local geometric and combinatorial lemmas while leaving the global arrangement extraction visibly unfinished.

The accompanying repository [29] records source revisions, negative searches, axiom reports, and the boundary between proved statements and proposed research. This separation is part of the result: a complete formal proof of a modest uniform improvement is stronger evidence than an ambitious recurrence with an unproved geometric step. It does not by itself confer numerical priority or solve the unrestricted Kobon problem.

3 Real-line geometry and exact certificates

3.1 The two extremal problems

Definition 3.1. An affine line is a set $\{(x, y) \in \mathbb{R}^2 : ax + by = c\}$ with $(a, b) \neq (0, 0)$. An arrangement is a finite set of distinct lines. A triangular cell is a bounded connected

component of the complement whose closure is a nondegenerate triangle. The maximum number of such cells among arrangements of n lines is $K(n)$. The maximum among simple arrangements, with neither parallel lines nor triple concurrence, is $K_s(n)$.

The classical problem permits degeneracies that the simple problem excludes. Consequently $K_s(n) \leq K(n)$, and a simple lower-bound construction is valid for both problems. The reverse use of an upper bound is invalid without a separate reduction. Our coordinate validator permits several lines through a triangular vertex, but excludes every line that cuts the triangle's interior.

The formal lower-bound predicate uses witnesses with no parallel pairs. This is a sufficient class of witnesses, not a claim that every maximizer of the classical problem has that property. For nontrivial orders, pairwise nonzero determinants also imply that every line is valid and that no two supporting lines coincide. The cases with fewer than three lines are handled separately.

3.2 A polynomial sign certificate

Represent a line ℓ by (a_ℓ, b_ℓ, c_ℓ) , with affine evaluation $f_\ell(x, y) = a_\ell x + b_\ell y - c_\ell$. For two lines set

$$D(\ell, m) = a_\ell b_m - b_\ell a_m, \quad (8)$$

$$\begin{aligned} V(\ell, m) &= (X_{\ell m}, Y_{\ell m}, D(\ell, m)) \\ &= (c_\ell b_m - b_\ell c_m, a_\ell c_m - c_\ell a_m, D(\ell, m)). \end{aligned} \quad (9)$$

When $D(\ell, m) \neq 0$, their actual affine intersection is $p_{\ell m} = (X_{\ell m}/D(\ell, m), Y_{\ell m}/D(\ell, m))$. For a third line r define

$$E(r; \ell, m) = a_r X_{\ell m} + b_r Y_{\ell m} - c_r D(\ell, m), \quad (10)$$

$$O(r; \ell, m) = E(r; \ell, m) D(\ell, m). \quad (11)$$

The useful identity is

$$O(r; \ell, m) = f_r(p_{\ell m}) D(\ell, m)^2. \quad (12)$$

Thus O has the sign of the affine evaluation without requiring division in the certificate. All operations are polynomial over integer or rational coordinates.

Definition 3.2 (Certified supporting triple). For a pairwise nonparallel arrangement $(\ell_0, \dots, \ell_{n-1})$, an increasing triple $i < j < k < n$ is certified if

1. $E(\ell_k; \ell_i, \ell_j) \neq 0$;
2. for each line r of the arrangement, the three numbers

$$O(r; \ell_i, \ell_j), \quad O(r; \ell_i, \ell_k), \quad O(r; \ell_j, \ell_k)$$

are either all nonnegative or all nonpositive.

The first condition excludes concurrence of the three supports; the second permits zeros at vertices but forbids a strict sign change.

Theorem 3.3 (From a sign certificate to a triangular cell). *Every certified supporting triple determines a nondegenerate bounded triangular cell. Its interior is nonempty. Two distinct increasing certified triples determine disjoint interiors.*

Proof. The three pair intersections exist because their denominators in (8) are nonzero. The nonvanishing evaluation excludes concurrence and gives three noncollinear points. Their positive barycentric combinations form a nonempty open triangle Δ .

By (12), each arrangement line has a common weak sign at the three vertices. Affine linearity gives the same sign throughout the closed triangle. It is strict in Δ : otherwise all three vertex evaluations would be zero, forcing a valid line to contain three noncollinear points. Hence no arrangement line enters Δ .

Fix an interior point z . Its strict sign cell is

$$S_z = \{x \in \mathbb{R}^2 : f_r(x)f_r(z) > 0 \text{ for every arrangement line } r\}.$$

The preceding argument gives $\Delta \subseteq S_z$. Conversely, the three support inequalities alone force all three barycentric coordinates of x to be positive, so $S_z \subseteq \Delta$. The sign cell is an intersection of open half-planes and is convex; it is exactly a component of the complement. It is bounded by the three finite segments of the triangle.

If two certified interiors meet, they have the same strict sign vector and hence the same sign cell. Its three supporting lines are determined by its three open sides. Since distinct arrangement lines cannot share an open side, the sets of supports agree, and sorting their indices makes the triples equal. The contrapositive proves disjointness. Q.E.D.

The modules `Kobon.Euclidean` and `Kobon.Cells` prove the sign-cell and nonoverlap content through `interior_nonempty`, `interior_eq_signCell`, and `distinct_interiors_disjoint`. The connected-component interpretation is the elementary convexity argument above; it is not separately encoded through a topological component API. Thus the certificates have an explicit interpretation as actual Kobon cells.

3.3 What a finite lower-bound certificate establishes

Proposition 3.4 (Soundness of exact finite validation). *Let A be an array of n integer line coefficients, and let $\mathcal{T}$ be a list of at least T certified increasing triples with no duplicates. If all pair determinants are nonzero, then $K(n) \geq T$. If all increasing triples of lines additionally have nonzero concurrence determinant, then $K_s(n) \geq T$.*

Proof. The coefficient inclusion $\mathbb{Z} \hookrightarrow \mathbb{R}$ preserves polynomial equalities, strict inequalities, and weak signs. The resulting real arrangement therefore satisfies Definition 3.2. Apply Theorem 3.3 to the distinct triples. The additional concurrence test is exactly the simplicity condition. Q.E.D.

Rational coordinates are cleared of denominators line by line before the integer check. The mathematical lower bound needs only a list of T valid triangles; it need not prove that no further triangle exists in the same arrangement. For the catalog, however, two independent exact counters also enumerate the triangle count. One works from consecutive intersections along each line, while the other directly tests triples against the arrangement. Agreement is an independent check on extraction and counting, not a substitute for the general soundness theorem.

These distinctions explain the three evidential levels used later. A proved symbolic theorem applies at arbitrary order. A finite Lean certificate establishes its listed order. An exact experimental count outside the catalog remains a computationally verified witness until promoted through the same formal pipeline. No finite list, however large, proves an infinite family.

4 A parity-sensitive all-order construction

This section gives the mathematical ingredients of Theorem 1.1. The underlying trigonometric family is a version of the classical Füredi–Palásti family. Its attribution is independent of the present formalization and choice of affine chart.

4.1 Trigonometric geometry and the classical count

For $0 < \theta < \pi$, define

$$\ell(\theta) : \sin \theta x + \cos \theta y = \sin(3\theta). \quad (13)$$

The normal has length one. Direct trigonometric expansion gives

$$D(\ell(x), \ell(y)) = \sin(x - y), \quad (14)$$

$$E(\ell(z); \ell(x), \ell(y)) = 4 \sin(x - y) \sin(z - y) \sin(z - x) \sin(x + y + z). \quad (15)$$

In particular, distinct angles in $(0, \pi)$ give nonparallel lines. Multiplying (15) by (14) yields

$$O(\ell(z); \ell(x), \ell(y)) = 4 \sin^2(x - y) \sin(z - y) \sin(z - x) \sin(x + y + z). \quad (16)$$

Only three factors determine the sign on the right.

For the half-phase choice $\theta_i = (i + 1/2)\pi/n$, select increasing triples with

$$i + j + k \equiv n - 1 \text{ or } n - 2 \pmod{n}. \quad (17)$$

For the shifted choice $\theta_i = (i + 1/6)\pi/n$, select instead

$$i + j + k \equiv 0 \text{ or } n - 1 \pmod{n}. \quad (18)$$

Lemma 4.1 (Triangle selection). *For $n \geq 3$, the triples in (17) certify triangles of the half-phase arrangement. The triples in (18) certify triangles of the shifted arrangement. Both arrangements are simple.*

Sign proof. The angle sum of any three lines is $(i + j + k + 3/2)\pi/n$ in the first arrangement and $(i + j + k + 1/2)\pi/n$ in the second. It is never an integer multiple of π , which, together with (15), excludes triple concurrence.

For a selected $i < j < k$, fix a fourth index r . The factors $\sin(\theta_r - \theta_j)$ and $\sin(\theta_r - \theta_i)$ change sign only when r passes the corresponding support index. For each of the pairs (i, j) , (i, k) , (j, k) , the remaining factor is $\sin((i + j + r + 3a)\pi/n)$ with $a = 1/2$ or $1/6$. Substituting the selected residue of $i + j + k$ fixes on which side of the next multiple of π this sum lies when $r < k$ and when $r > k$. Splitting the index range at i, j, k shows that the three expressions in (16) have one weak sign in each range. At a support index the appropriate evaluations are zero. Their common sign need not be the same for different r ; that is precisely the permitted sign choice in Definition 3.2.

Equivalently, this check uses only the sign of $\sin t$ on consecutive intervals $(s\pi, (s + 1)\pi)$ and the inequalities $0 \leq r < n$. No numerical trigonometric approximation is used. The complete index splits are proved in `FurediPalasti.lean` and `ShiftedFurediPalasti.lean`. Definition 3.2 and Theorem 3.3 then give the claimed cells. Q.E.D.

For any residue $b \in \mathbb{Z}/n\mathbb{Z}$, the equation $i + j + k = b$ has exactly n^2 ordered solutions. At most $3n$ have a repeated index, because each of $i = j$, $j = k$ and $i = k$ leaves at most n choices. Two distinct residue classes therefore give at least $2n^2 - 6n$ ordered distinct-index

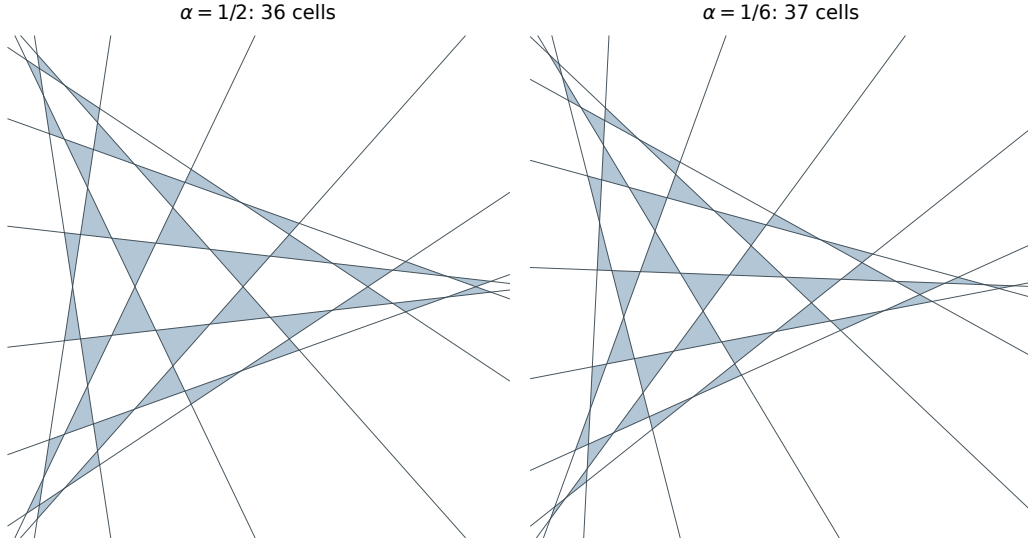


Figure 2: New rational-coordinate illustrations at $n = 12$, obtained from the half-phase and shifted trigonometric families. Two independent exact counters certify 36 and 37 triangles in these rational proxies. The axes of each panel are rescaled independently. The real-coordinate universal theorem follows from the symbolic sign and modular arguments, independently of this illustration. The underlying family is attributed to Füredi and Palásti.

triples. Each increasing triple accounts for six permutations, so the half-phase arrangement has at least

$$\left\lceil \frac{2n^2 - 6n}{6} \right\rceil = B(n) \quad (19)$$

certified triangles. This is the classical all-order estimate, now proved directly in the real-coordinate formalization.

4.2 The repeated-index correction

Lemma 4.2 (Shifted count). *For every $n \geq 3$, the shifted arrangement has at least*

$$\left\lfloor \frac{n(n-3)}{3} \right\rfloor + 1$$

certified triangular cells.

Proof. Use the two residues 0 and $n-1$ in (18). In the zero residue, the three repeated-index sets all contain $(0, 0, 0)$. Their union therefore has at most $3n-2$ members, not merely $3n$. In the other residue retain the bound $3n$. The number of ordered distinct-index solutions is at least $2n^2 - 6n + 2$. If T is the number of increasing selected triples, then

$$6T \geq 2n^2 - 6n + 2, \quad T \geq \left\lceil \frac{n(n-3) + 1}{3} \right\rceil = \left\lfloor \frac{n(n-3)}{3} \right\rfloor + 1.$$

Lemma 4.1 supplies their actual geometry. This argument uses the overlap of bad-index sets, not a numerical rounding conjecture. Q.E.D.

4.3 Projective changes of affine chart

The shifted count is enough for the even part of (1). To obtain the additional odd-order gain we use exposed vertices of the half-phase arrangement. The exact coordinate map is

$$(x, y) \mapsto \left(\frac{x}{h-x}, \frac{y}{h-x} \right), \quad h \neq 0. \quad (20)$$

It replaces the line $x = h$ by the line at infinity. The transformed coefficients of $ax + by = c$ are

$$\Phi_h(a, b, c) = (ha - c, hb, c). \quad (21)$$

Unlike an affine transformation, this change can alter which projective triangular faces appear bounded. Thus counting triangles before and after the change requires an explicit sign argument.

Lemma 4.3 (Exact covariance). *If $D(\ell, m) \neq 0$, then*

$$D(\Phi_h \ell, \Phi_h m) = hD(\ell, m)(h - x_{\ell m}), \quad (22)$$

$$E(\Phi_h r; \Phi_h \ell, \Phi_h m) = h^2 E(r; \ell, m), \quad (23)$$

$$O(\Phi_h r; \Phi_h \ell, \Phi_h m) = h^3 (h - x_{\ell m}) O(r; \ell, m), \quad (24)$$

where $x_{\ell m}$ is the abscissa of the old intersection. If the cut avoids every old vertex, the transformed arrangement has no parallel pairs. Triple nonconcurrency is preserved.

Proof. Substitute (21) into the polynomial expressions in Section 3, expand, and cancel. Nonzero h and avoidance of all $x_{\ell m}$ give the pair determinant claim, while multiplication by h^2 preserves a nonzero triple evaluation. Q.E.D.

The sign multiplier in (24) also gives a direct triangle-preservation rule. If $h > 0$ and all vertices of a certified triangle lie to the left of $x = h$, all three multipliers are positive. If $h < 0$ and its vertices lie to the right, they are again positive. At selected exposed vertices on the opposite side, the sign reverses. Carefully choosing which vertices are separated by the cut creates the extra triangles.

Lemma 4.4 (One exposed vertex). *For odd $n = 2m + 1 \geq 3$, a change of affine chart of the half-phase arrangement has at least $B(n) + 1$ bounded triangles.*

Proof. The intersection of $\ell(x)$ and $\ell(y)$ has abscissa

$$X(x, y) = \cos(2x) + \cos(2y) + \cos(2x + 2y). \quad (25)$$

For the half-phase angles, the vertex of lines $0, n - 1$ is uniquely rightmost, with abscissa $c_n = 1 + 2 \cos(\pi/n) > 0$. Indeed, each of the first two cosine terms in (25) is at most $\cos(\pi/n)$ and the last is at most one; if the pair is not $(0, n - 1)$, at least one necessary inequality is strict. Choose $h > 0$ between all other abscissae and c_n .

No selected triangle of (17) uses the pair $(0, n - 1)$: such a triple would require its middle index to be 0 or $n - 1$ modulo n . Hence all counted triangles are preserved by the positive multiplier rule. The additional triple $(0, m, 2m)$ has, for each test line, nonnegative evaluation at the exposed vertex and nonpositive evaluations at its other two vertices before the transformation. The cut reverses exactly the exposed-vertex sign, so the transformed triple satisfies the certificate. These signs follow from (16) by splitting the test index at m . The triple is absent from the old selected list, and Lemma 4.3 preserves simplicity. This gives one additional bounded triangle. Q.E.D.

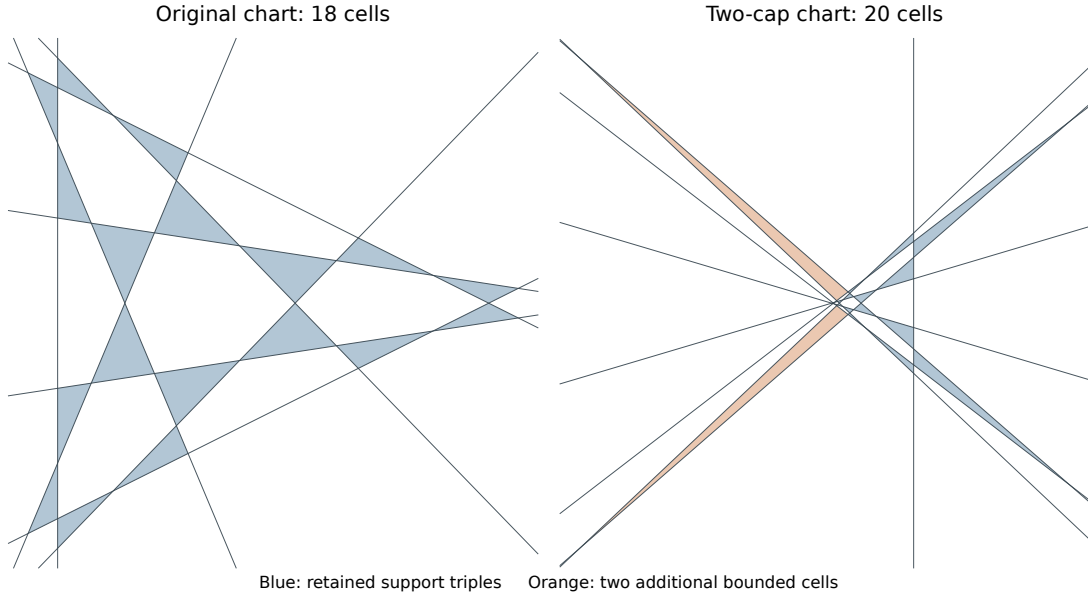


Figure 3: New exact rational illustration at $n = 9$: the change of affine chart retains all 18 old triangular cells and creates two more. The resulting 20 is below the known optimum 21; this figure illustrates the uniform mechanism. Panels are rescaled independently. Both counts are checked exactly. The general proof tracks the determinant multiplier at every vertex, independently of the drawing.

Lemma 4.5 (Two exposed vertices). *For $n = 6k + 3$ with $k \geq 1$, a change of affine chart of the half-phase arrangement has at least $B(n) + 2$ bounded triangles.*

Proof. Write $n = 3m$, where $m = 2k + 1$. The two intersections supported by $(m - 1, m)$ and $(2m - 1, 2m)$ have common abscissa $-c_n/2$. Every other intersection has strictly larger abscissa. To verify the latter assertion, rewrite (25) as

$$X(u, v) = 2zw + 2z^2 - 1, \quad z = \cos(u + v), \quad w = \cos(u - v).$$

For nonadjacent angle indices, the cosine bounds on the discrete grid, together with completing the square in z , give $X > -c_n/2$. For adjacent indices, $w = \cos(\pi/n)$ and the same quadratic attains equality precisely at $u + v = 2\pi/3$ or $4\pi/3$, giving the two listed pairs. The exceptional wrapping pair $(0, n - 1)$ is the rightmost vertex. These exposure inequalities are proved uniformly in `FurediPalastiCaps.lean`.

Choose $h < 0$ strictly above $-c_n/2$ and below all other abscissae. The modular conditions (17) show that no counted triangle uses either exposed pair. Its vertices are therefore all to the right of the cut and its signs are preserved. The two additional support triples are

$$(2k, 2k + 1, 5k + 2), \quad (k, 4k + 1, 4k + 2).$$

For each test index, substitution into (16) gives one sign at the exposed pair and its opposite weak sign at the other two vertices. The negative cut reverses exactly the exposed-vertex multiplier in each case, establishing the transformed certificates. The two triples are distinct and neither satisfies the old selection residues. Simplicity follows from Lemma 4.3. Appending both triples to the retained list proves the claim. Q.E.D.

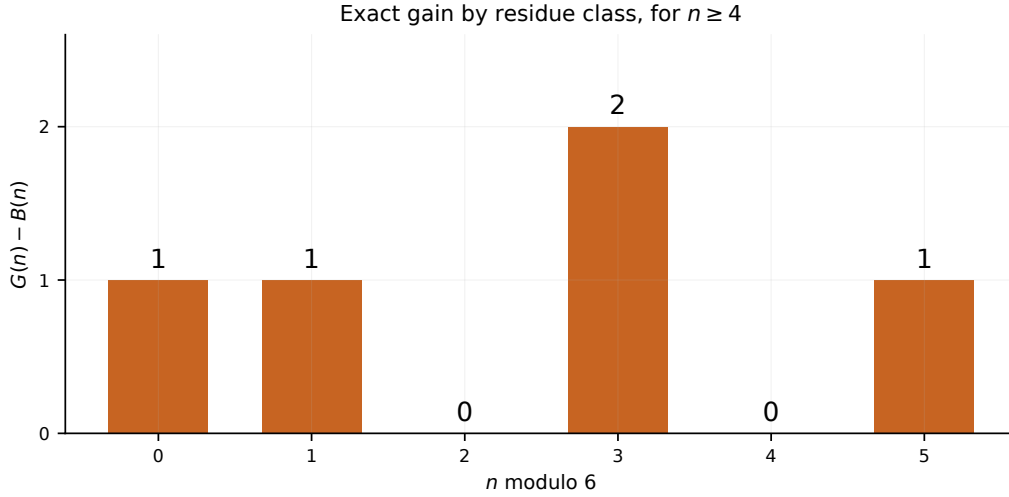


Figure 4: The exact proved gain $G - B$, resolved by residue class modulo six. The gain is strict on four classes; the universal statement includes arbitrarily large orders in each class.

4.4 Assembly, exact gain, and asymptotic scale

Proof of Theorem 1.1. Use Lemma 4.2 for even $n \geq 4$. For odd n not divisible by three, $n(n-3) \equiv 1 \pmod{3}$, so $B(n) + 1 = \lfloor n(n-3)/3 \rfloor + 2$; apply Lemma 4.4. For odd multiples of three at least nine, $B(n) = n(n-3)/3$ and Lemma 4.5 gives the same target. Three nonconcurrent lines give $G(3) = 1$, while the smaller cases have zero triangles. Each construction used is simple. Q.E.D.

Corollary 4.6 (Exact improvement over the quoted baseline). *For $n \geq 4$,*

$$G(n) - B(n) = \begin{cases} 1 + (n \bmod 2), & 3 \mid n, \\ n \bmod 2, & 3 \nmid n. \end{cases}$$

The gains in classes $0, 1, 2, 3, 4, 5 \pmod{6}$ are respectively $1, 1, 0, 2, 0, 1$.

Proof. The product $n(n-3)$ is zero modulo three when $3 \mid n$, and one modulo three otherwise. Substitute these remainders in the floor and ceiling formulas. Q.E.D.

For comparison with the classical polynomial benchmark $U(n) = \lfloor n(n-2)/3 \rfloor$, direct integer arithmetic gives, for $n \geq 4$,

$$U(n) - G(n) = \left\lfloor \frac{n}{3} \right\rfloor + \mathbf{1}_{n \equiv 2 \pmod{3}} - 1 - (n \bmod 2). \quad (26)$$

This comparison does not formalize the published upper-bound theorem. It quantifies the scale of the remaining gap to that benchmark. It is still of order n , and stronger upper bounds or special constructions apply at some orders.

The outcome is an unconditional affine theorem, stronger than the quoted baseline at infinitely many orders. Its numerical first priority remains a separate literature question. The construction gives a witness for each n ; it does not provide an operation that adds one line to every preceding optimal witness.

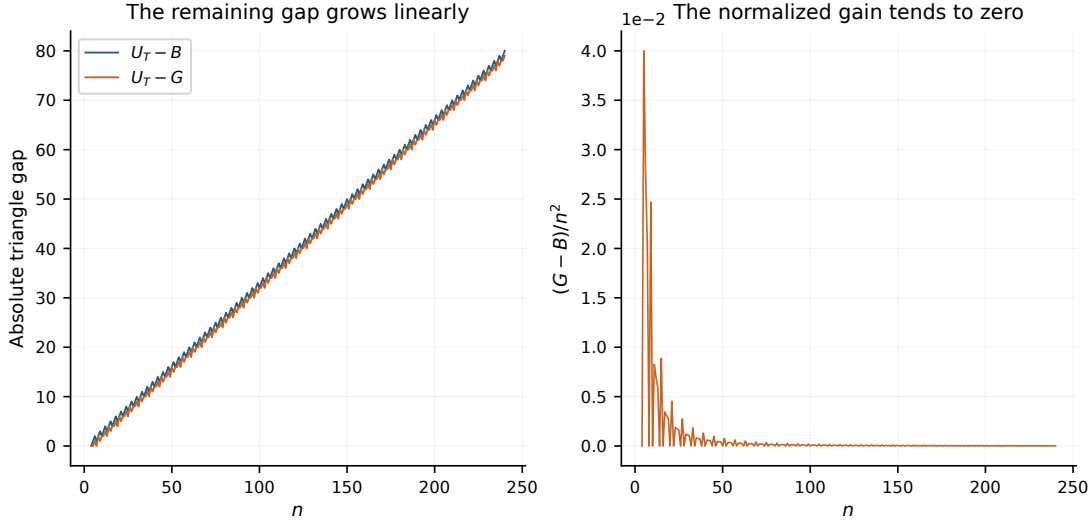


Figure 5: Deficit and normalized growth computed from the proved formulas. The constant improvement is visible in the deficit, while both general lower formulas have the same leading quadratic and linear terms. The polynomial used in the deficit is a literature benchmark, not a new upper theorem of this paper.

5 Individual cases, finite enhancements, and provenance

5.1 The March values and their present verification

The three finite inequalities associated with the author’s early extension work are

$$K(28) \geq 238, \quad K(30) \geq 275, \quad K(34) \geq 357. \quad (27)$$

Each now has an exact simple-arrangement certificate, so the same inequalities hold for K_s . Their present validity is independent of the unproved recurrence in the original March manuscript. The certificates give actual lines and complete checked triangle lists, not only numerical consequences of an extension hypothesis.

The current OEIS table associates the name Zarzuelo with these entries [17, 28]. Attribution in an index and first discovery are different questions. The earlier literature already contains matching constructions or consequences for some of these counts; the literature review records these overlaps. The claims made here are the documented attribution trail, the independently verified witnesses, and their place in the author’s extension program. No new priority claim is inferred from the absence of a value from a particular online table.

The original inventory also listed outputs for every order from 3 through 60, together with larger cases at 99 and 195. Some were direct certificates, some were numerical consequences of a boundary-defect argument, and many were weaker than an already available construction. Appendix C preserves that inventory as history. The current certificate aliases independently retain its numerical inequalities; a label inherited from the old table is not used as a present formal-proof status.

5.2 Materialized improvements from the phase and chart construction

Table 4 records simple rational witnesses promoted during the last construction phase. Each listed new count is supplied by the universal theorem and also realized by an independently checked finite coordinate file. This double presentation is useful: the theorem proves all orders, while the coordinates support visualization and further exact searches.

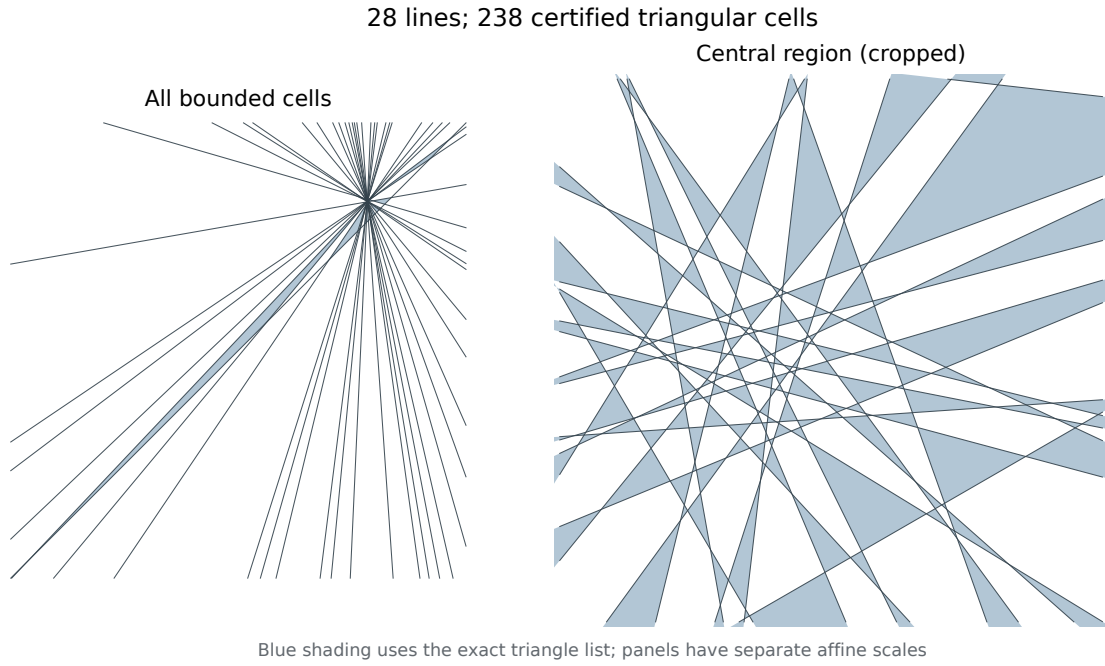


Figure 6: The retained exact 28-line witness with 238 certified triangles. This is a new rendering of the saved coordinates associated with the earlier extension program. The marked zoom, where present, is an enlargement of the same arrangement; it does not add or remove lines from the certificate. The picture illustrates a lower bound, while the exact sign tests establish it.

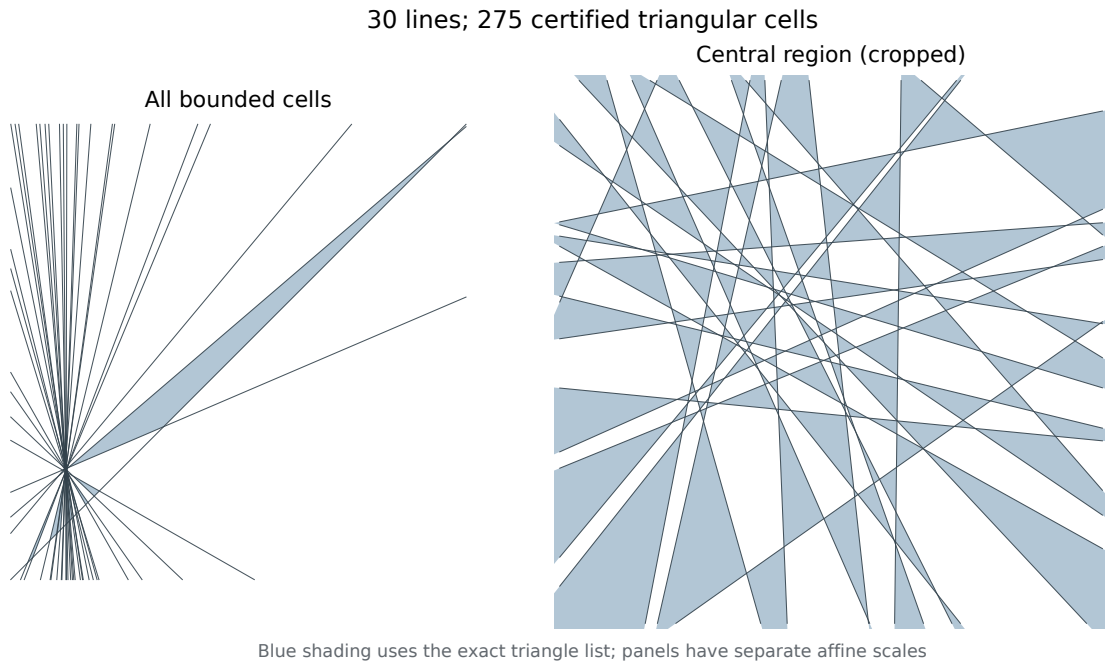


Figure 7: The retained 30-line witness with 275 certified triangles, rendered from its exact coordinates. The count has prior literature overlap and is not asserted to be a new numerical record. The complete witness remains in the catalog even though the historical general extension proof was incomplete.

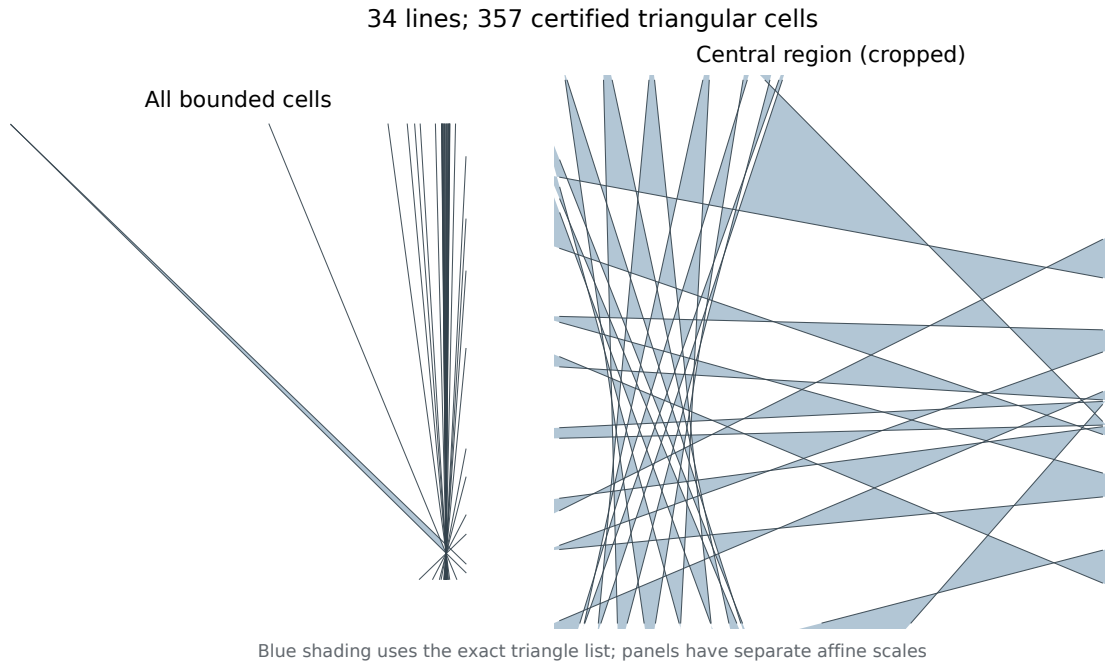


Figure 8: The retained 34-line witness with 357 certified triangles. Original plotting of the certificate makes the finite conclusion inspectable independently of the claimed universal extension. The construction input and earlier matching literature are credited separately.

n	Earlier retained count	Current simple count	Increase
39	469	470	1
47	690	691	1
48	720	721	1
51	817	818	1
53	884	885	1
54	918	919	1
55	954	955	1
59	1102	1103	1
60	1140	1141	1
99	3169	3170	1
195	12481	12482	1

Table 4: Changes relative to this project’s previous retained witnesses, not a table of first discoveries or global record improvements. “Simple” is certified for every current coordinate witness in this table.

In particular, the earlier additional-line values 39 : 469, 51 : 817, 99 : 3169 and 195 : 12481 are now exceeded by G itself. Their finite correctness survives, but those numbers are not distinctive of the additional-line family. Historical evidence is retained rather than deleted when a stronger construction is found.

5.3 The 44-line simple case

The exact exterior extension of the retained 43-line arrangement creates 21 triangles while preserving its 587 old triangles. Both exact counters and the Lean certificate give a simple 44-line arrangement with 608 triangles. The published simple upper bound of Blanc at this order is also 608 [2]. Thus, using that external upper theorem,

$$K_s(44) = 608. \quad (28)$$

The repository proves the lower inequality. It does not formally prove Blanc’s theorem, and (28) does not establish $K(44) = 608$ in the unrestricted classical problem. Nor does materializing this consequence establish first discovery of the numerical value.

This example shows why upper-bound scope belongs next to the numerical statement. A simple witness can establish a classical lower bound, but a simple upper theorem cannot by itself close the classical problem.

5.4 External records and independently checked inputs

The updated catalog incorporates exact gallery witnesses from Parpalak and Utkin at

8 : 15, 14 : 54, 20 : 117, 26 : 204, 32 : 315, 36 : 402, 38 : 450, 42 : 553, 46 : 667, 50 : 792.

These are constructions or coordinate sources of the cited researchers, not newly discovered values in this project. The eight-line value has older priority; the modern gallery is the source of the checked coordinates. The public source snapshot and the formal certificate index preserve the provenance [17, 18].

All fifteen of Maiorana’s supplied 14-line arrangements with 54 triangles were independently checked and promoted as finite certificates [16]. Fourteen have two triple points and one has four. Their source license and immutable commit are retained. These witnesses are a particularly useful warning against conflating simple and nonsimple arrangements: removing a multiple intersection by a small perturbation need not preserve the full triangle count. Section 9 reports the exact resolution experiments.

The two source snapshots used for these imports are Maiorana commit `e47c7cfd9661e54d29befb83118bf83d5f804028` and gallery commit `feae4f5571a988a3ed45bd26a36259ebb7d7d3d2`. Their mathematical attribution is attached to each source record even when a new Lean wrapper or new figure is generated here.

5.5 The finite envelope and the complete tables

Proposition 5.1 (Retained all-order envelope). *Let $C(n)$ be the maximum count in the finite catalog at order n , or zero if that order is absent. Then $K(n) \geq L(n) = \max\{G(n), C(n)\}$ for every n . In the frozen release, $L(n) = G(n)$ for every $n > 195$.*

Proof. When $C(n) \leq G(n)$, use Theorem 1.1. Otherwise choose a certificate attaining the finite maximum and apply Proposition 3.4. No coordinate certificate in this release has order greater than 195. Larger computational witnesses discussed later have not been silently added to the Lean catalog. Q.E.D.

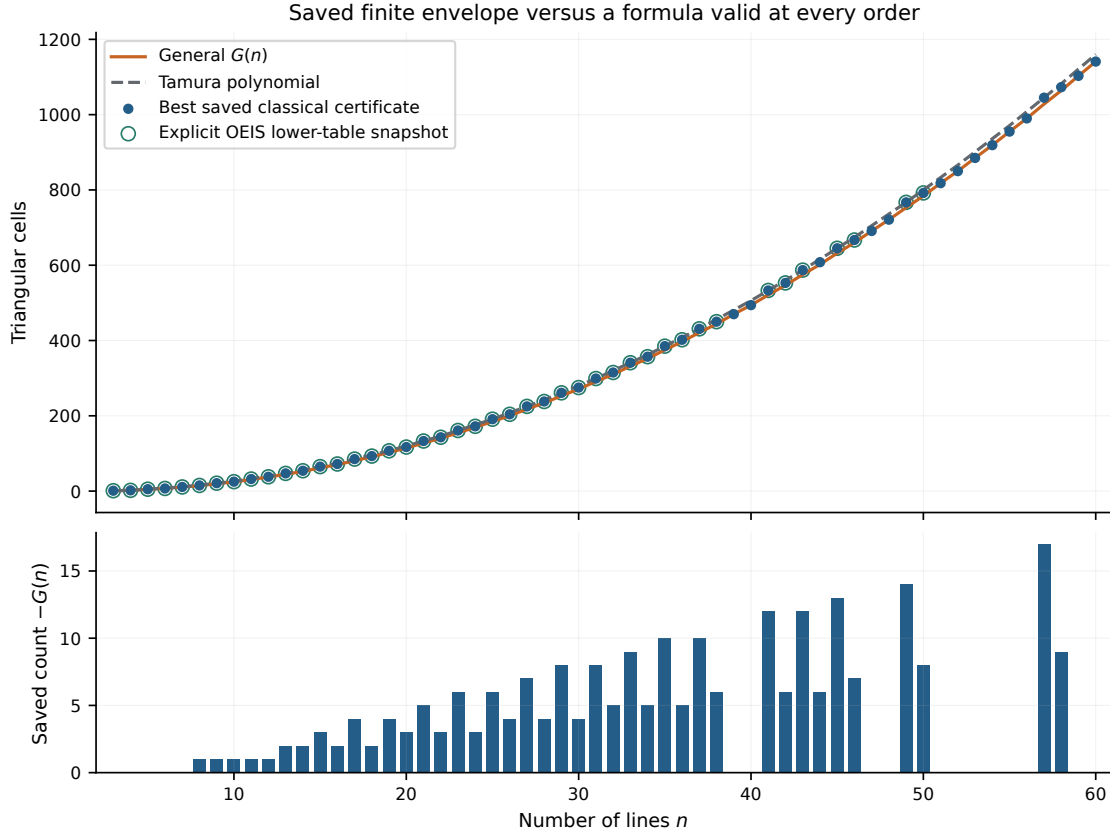


Figure 9: The strongest saved simple and classical coordinate counts, compared with the uniform formula and the polynomial upper benchmark. External witnesses retain their attribution. A marker above G is a finite enhancement of the general formula, not automatically an original record.

The formal definition is `Universal.bound = max Universal.baseline AllN.bound`. Since $G \geq B$, it equals the expression in Proposition 5.1; the older AllN layer preserves the previous finite-envelope theorem. This makes comparisons reproducible without rewriting the historical construction history.

Appendix A gives the best saved simple and classical values from 3 through 60. Appendix B lists all 138 certificate identities, including repeated orders and weaker checkpoints. Appendix C gives the older author-output inventory. Together these tables prevent three common confusions: a retained output is not necessarily the best available value; a missing OEIS row is not proof of novelty; and an exact count for one arrangement is not automatically the extremal value $K(n)$.

6 Exterior extensions: the March proposal and its precise replacement

The starting point of this project was the March 2026 proposal that an odd Kobon arrangement could always be extended with the gain

$$K(2m+2) \geq K(2m+1) + m. \quad (29)$$

The proposal was presented in [28] and subsequently described as an iterative rule in [13]. The preserved March source is important evidence of the development of the idea, but is not a

completed proof of (29): its Lean file contains eight proof placeholders, including the extension assertion. The present work replaces the informal alternating-segment argument by an actual geometric extension theorem and identifies exactly which additional counting hypothesis yields the desired gain. Independently verified finite bounds are retained regardless of the status of that original argument.

The later OEIS attribution at orders 28, 30, and 34 is evidence of the proposal's reception, rather than a proof of numerical priority [17]. In particular, the BBL table already records the straight-line value $30 : 275$. The perfect 33-line input belongs to the family of Forge and Ramírez Alfonsín [8]; applying the exterior extension stated by BBL gives $34 : 357$ [1]. The historical attribution, the validity of a finite count, and the novelty of a general extension theorem are therefore assessed separately.

Two distinctions are essential. A recurrence for maxima need not extend *every* arrangement attaining a given count, and a construction for one odd-to-even step need not admit repeated full-gain extensions. Throughout this section the geometric hypotheses concern simple affine arrangements; consequently their conclusions give lower bounds for both K_s and K , but nonsimple witnesses for K cannot automatically be substituted into the hypotheses.

6.1 A geometric theorem from explicitly visible pairs

Write an old line as $L_i = \{p : \ell_i(p) = 0\}$, with ℓ_i affine, and let $A = \{L_0, \dots, L_{n-1}\}$ be simple. A linear functional w is *admissible* if it is nonzero on every old line direction. At $p_{ij} = L_i \cap L_j$, choose the direction vectors u_i, u_j on the two supports so that $w(u_i) = w(u_j) = 1$. The following criterion uses only the old arrangement.

Definition 6.1 (Visible pair). The pair $\{i, j\}$ is visible from w if, for every old affine equation ℓ_k , the three real numbers

$$\ell_k(p_{ij}), \quad D\ell_k(u_i), \quad D\ell_k(u_j)$$

have one common weak sign: either all are nonnegative or all are nonpositive. Here $D\ell_k$ denotes the linear part of ℓ_k .

Geometrically, the two increasing- w rays bound an unbounded empty wedge. The algebraic definition is convenient because it gives the required half-plane inequalities directly, without assuming that a future triangle already exists.

Theorem 6.2 (Verified exterior realization). *Let A be a simple arrangement of n real affine lines, let $\mathcal{T}$ be a set of T distinct triangular cells of A , and let $\mathcal{V}$ be a set of c distinct pairs visible from an admissible w . If*

$$H > \max_{i < j} w(p_{ij}),$$

then adjoining $L = \{p : w(p) = H\}$ produces a simple arrangement containing all triangles in $\mathcal{T}$ and at least c further triangles. In particular, $K_s(n+1) \geq T + c$.

Proof. Admissibility excludes a parallel old line, and the strict height inequality excludes every old vertex from L . Thus simplicity is preserved. Every old triangle lies in the convex hull of its vertices, strictly below the level H , and survives unchanged.

For a visible pair, the new vertices on its two supports are

$$p_{ij} + (H - w(p_{ij}))u_i, \quad p_{ij} + (H - w(p_{ij}))u_j.$$

At either vertex an old affine equation has the form

$$\ell_k(p_{ij}) + (H - w(p_{ij}))D\ell_k(u).$$

The second coefficient is positive. Definition 6.1 therefore puts all three vertices in one closed half-plane of each old line. The new line is itself a supporting side, and the triangle is nondegenerate because the old intersection lies strictly below it. No arrangement line cuts its interior. Different old pairs give different support triples, and every new triple uses L , whereas no old one does. The two lists are consequently disjoint. Q.E.D.

This is the geometric content of **Exterior.extension**. The theorem is proved for arbitrary real coefficients and either parity, with no assumption about a future triangle count. An explicit formal height is

$$H = 1 + \sum_{\substack{i,j < n \\ i \neq j}} |w(p_{ij})|,$$

so existence of a sufficiently distant line is part of the construction. The proof uses the standard Lean logical axioms and no finite native evaluation. What it does *not* supply is a universal lower bound of $\lfloor n/2 \rfloor$ on the number of visible pairs.

6.2 Defect, boundary wedges, and a rule for both parities

Let $t(A) = T$ and define the bounded-segment defect

$$d(A) = n(n - 2) - 3T. \quad (30)$$

There are $n(n - 2)$ elementary bounded segments. A segment can support at most one triangular cell: its endpoints determine the two other supports, and their intersection lies on at most one side of the segment. Thus $d(A)$ counts the unused bounded segments. Let $b(A)$ count the unbounded two-sided cells, called wedges below.

Lemma 6.3 (Boundary charging). *For a simple arrangement of $n \geq 3$ lines,*

$$3 \leq b(A) \leq n, \quad b(A) + d(A) \geq n.$$

Proof. The arrangement has $2n$ free rays. A wedge pairs two such rays at their common endpoint, and each free ray belongs to at most one wedge. An unpaired ray on L , ending at $p = L \cap M$, meets an interior vertex of M . The two bounded elementary segments of M incident to p cannot both support triangles: both triangles would also use the unique bounded segment of L starting at p . Charge the unpaired ray to an unused segment of M at p . Each unused segment receives at most two charges, one at each endpoint. Hence $2n - 2b \leq 2d$. This also proves $b \leq n$. Finally, the convex hull of all finite vertices has at least three vertices. Both supports at a hull vertex have free outgoing rays, which give a wedge. Distinct hull vertices give distinct wedges. Q.E.D.

Order the $2n$ oriented line directions cyclically. The two directions of a wedge are consecutive: a line direction strictly inside that sector would force a line to enter the wedge and cross a free boundary ray. Write $B \subset \mathbb{Z}/(2n)$ for the wedge-start indices. Define

$$c_j = |B \cap \{j, j + 1, \dots, j + n - 2\}|, \quad C(A) = \max_j c_j. \quad (31)$$

The n directions on which an admissible w is positive form one consecutive block, and every such block occurs for some normal sector. An exterior line closes precisely the wedges whose two rays lie in this block. Consequently $C(A)$ is the exact largest exterior gain and

$$\sum_{j=0}^{2n-1} c_j = (n - 1)b(A), \quad \left\lceil \frac{(n - 1)b(A)}{2n} \right\rceil \leq C(A) \leq \left\lfloor \frac{n}{2} \right\rfloor. \quad (32)$$

The sum follows by counting the $n - 1$ blocks containing a fixed wedge. The upper bound follows because complete wedge pairs use disjoint rays in a block of n rays.

Theorem 6.4 (Boundary-defect extension). *For a simple n -line arrangement with $n \geq 3$ and T triangles, an exterior line preserves the old triangles and gives at least*

$$T + g(n, T), \quad g(n, T) = \left\lceil \frac{n-1}{2n} \max\{3, 3T - n(n-3)\} \right\rceil \quad (33)$$

triangles. Thus, for either parity,

$$K_s(n) \geq T \implies K_s(n+1) \geq T + g(n, T).$$

Proof. Lemma 6.3 gives $b(A) \geq \max\{3, n-d(A)\}$. Substitute into (32) and apply Theorem 6.2. If T is only a lower bound, the witness has some count $T' \geq T$; the right side of (33) is nondecreasing in T , which gives the stated maximum-function implication. Q.E.D.

Remark 6.5 (Formal scope and attribution). Theorem 6.4 is an ordinary geometric theorem in this manuscript. The cyclic sum, averaging argument, and defect arithmetic are Lean checked, and Theorem 6.2 supplies actual geometric realization. The universal charging map and the correspondence between Euclidean boundary rays and the finite cyclic data have not yet been joined into an end-to-end Lean proof. Exterior addition and unused-segment arguments have precedents in [1, 2]. The contribution under investigation is the explicit quantitative defect formulation and its certificate interface; first numerical or historical priority is not asserted.

Corollary 6.6 (Full gain for an odd saturated input). *If a simple arrangement of $2m+1$ lines has defect at most two, an exterior line creates exactly m additional triangles. In particular, this applies to every simple odd arrangement attaining $\lfloor n(n-2)/3 \rfloor$.*

Proof. The wedge count is at least $2m - 1$. If all $4m + 2$ direction sectors had gain at most $m - 1$, then

$$2m(2m - 1) \leq 2mb(A) = \sum_j c_j \leq (4m + 2)(m - 1) = 2m(2m - 1) - 2,$$

a contradiction. Equation (32) bounds the gain above by m . The segment-count upper bound at odd n leaves defect zero or two. Q.E.D.

For an even input $n = 2m$, the sufficient condition $b(A) \geq 2m - 1$ likewise forces gain m , because $(2m - 1)^2 > 4m(m - 1)$. In either parity the exact criterion is the finite condition $C(A) = \lfloor n/2 \rfloor$. There is no claim that every even or odd maximum has this boundary profile. Iterating (33) is nevertheless legitimate ordinary geometry and supplies a weaker lower-bound rule from any simple seed.

6.3 What fails in indefinite full-gain exterior iteration

The March argument made the number of new triangles follow from an alternation assertion after placing a line far from all old vertices. Distance proves preservation; it does not prove emptiness for half the new segments. For example, the five lines $y = ix + i^2$, $i = 0, \dots, 4$, have three triangles, and adding $y = x/2 + 100$ creates only one further triangle. These counts are checked by exact certificates. This refutes that unqualified step of the argument, without refuting a better choice of line or an inequality about maxima.

A stronger obstruction concerns every chain of exterior additions. Write b_r for its wedge count and c_r for the gain at step r . Closing c_r wedges consumes those wedges. No new free

Input order	Triangles	Wedges	Best exterior gain	Requested gain
49	767	47	24	24
50	791	24	23	25
51	814	3	2	25
52	816	3	2	26
53	818	3	2	26

Table 5: One exact greedy exterior chain from the attributed 49-line input. Every direction sector was counted with rational arithmetic. These are capacities of the displayed arrangements, not upper bounds for K or K_s at the corresponding orders.

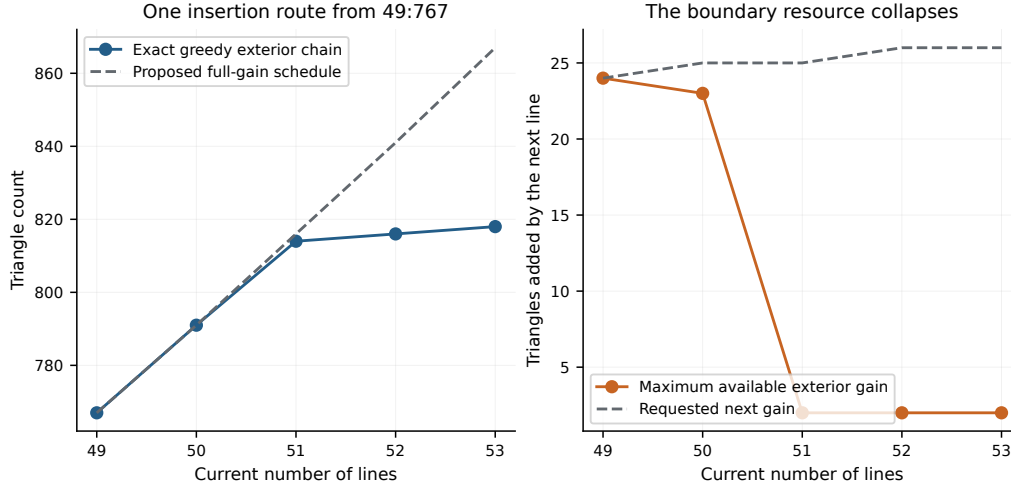


Figure 10: The boundary resource in the particular exact 49-line chain. The sharp drop in available wedges explains why its first full-gain extension cannot be repeated with the requested next gain. The plotted capacities concern this chain only.

ray can appear at an old vertex, and a wedge at a new vertex must use one of the two free rays of the newly added line. Therefore

$$b_{r+1} + c_r \leq b_r + 2, \quad \sum_{j < r} c_j \leq b_0 + 2r - b_r \leq b_0 + 2r - 3. \quad (34)$$

The accumulated exterior gain is at most linear in the number of steps. The accumulated gains $\lfloor n/2 \rfloor$ are quadratic, so an infinite chain of such full-gain exterior steps is impossible. The numerical contradiction is formalized in `Iteration.no_infinite_full_gain_budget`; as in Remark 6.5, the universal wedge-budget interpretation is an ordinary geometric argument.

Table 5 records the motivating test from 49 : 767. The obstruction is independent of the greedy tie break: two exterior gains 24 and 25 from this seed would imply

$$b_2 \leq 47 + 4 - 24 - 25 = 2,$$

contrary to $b_2 \geq 3$. The five finite profile maxima have a checked Lean arithmetic certificate; their identification with the saved coordinates is supplied by the exact geometry program. A reconstruction between steps or an interior insertion could evade this boundary budget, so it does not disprove the proposed recurrence for maxima.

6.4 A conditional recurrence and an unconditional numerical bound

For any integer-valued function F , assume

$$F(n+1) \geq F(n) + \lfloor n/2 \rfloor \quad (n \geq s). \quad (35)$$

Summation, including both parity cases, gives the fully checked arithmetic implication

$$F(n) \geq F(s) + \left\lfloor \frac{(n-1)^2}{4} \right\rfloor - \left\lfloor \frac{(s-1)^2}{4} \right\rfloor \quad (n \geq s). \quad (36)$$

The hypothesis (35) remains explicit in the formal theorem. Substituting $s = 49$ and $F(49) = 767$ yields the numerical target

$$Q_{49}(n) = \left\lfloor \frac{(n-1)^2}{4} \right\rfloor + 191. \quad (37)$$

There is now a separate unconditional Lean theorem

$$K(n) \geq Q_{49}(n) \quad (n \geq 49), \quad (38)$$

proved using the saved witnesses at 49 and 50 and the classical all-order baseline thereafter. Indeed, for $n \geq 51$,

$$\frac{n(n-3)}{3} - \frac{(n-1)^2}{4} - 191 = \frac{n^2 - 6n - 2295}{12} \geq 0,$$

with equality at $n = 51$ and an increasing numerator beyond it. Rounding gives $B(n) \geq Q_{49}(n)$. This proves the desired numerical formula without constructing a nested full-gain chain. The stronger all-order bound G and the finite envelope elsewhere in this paper also dominate it.

For comparison, direct iteration of the defect rule $T_{n+1} = T_n + g(n, T_n)$ from $49 : 767$ guarantees $50 : 791$, $51 : 803$, $52 : 805$, and $60 : 821$. The stronger target (37) gives $51 : 816$, $52 : 841$, and $60 : 1061$. These are different constructions and different assertions. The completed numerical theorem does not retrospectively prove the March extension claim, and the failure of its proposed exterior iteration does not invalidate the completed numerical theorem.

7 Compatible seeds, geometric doubling, and sparse families

The second extension mechanism inserts a whole collection of lines near a distinguished support. It belongs to Bartholdi, Blanc, and Loisel (BBL) [1]; it is fundamentally different from the one-line exterior operation of Section 6. The present work supplies explicit compatible seeds, an actual one-step geometric Lean proof, and finite certificates at considerably larger orders. The distinction between a published-theorem consequence and a completed recursive Lean theorem is maintained throughout.

7.1 The exact one-step theorem

Let $q = 4r$ with $r \geq 5$, put $\alpha = \pi/q$, and take

$$0 < \varepsilon < \tan(\alpha/2). \quad (39)$$

Define q ordered intercepts, indexed from zero, by

$$a_j = \begin{cases} \tan((j - q/2 + 1)\alpha), & 0 \leq j < q/2 - 1, \\ -\varepsilon, & j = q/2 - 1, \\ \varepsilon, & j = q/2, \\ \tan((j - q/2)\alpha), & q/2 < j < q. \end{cases} \quad (40)$$

In particular, the extreme intercepts are $-\cot \alpha$ and $\cot \alpha$. The two central intercepts replace the repeated zero that would otherwise occur in this grid.

Definition 7.1 (Compatible tangent-grid seed). A seed at (q, ε) consists of $Y_0 : y = 0$ and the q lines $L_j : y = m_j(x - a_j)$, with the following properties: the arrangement is simple; every interval $[a_j, a_{j+1}]$ on Y_0 supports the triangular cell with supports Y_0, L_j, L_{j+1} ; and the central apex $L_{q/2-1} \cap L_{q/2}$ lies above Y_0 . A counted seed additionally carries a list of distinct actual triangular cells whose length is at least T .

Simplicity already requires all m_j to be nonzero and pairwise distinct. The distinguished-line condition is a substantial geometric restriction, not a consequence of having a large total count. It is precisely why an arbitrary optimal arrangement cannot simply be used as a repeatable BBL seed.

Theorem 7.2 (Fully verified geometric doubling). *Suppose $q = 4r \geq 20$ and (39) holds. A compatible tangent-grid seed with at least T certified triangles gives a simple real straight-line arrangement of $2q + 1$ lines with at least*

$$T + q^2 \quad (41)$$

triangles.

The precise formal theorem is `BBLDoubling.doubling`. It assumes the actual old line equations, simplicity, each distinguished triangle, the positive central height, and a duplicate-free triangle list. It does not assume a future crossing order, a list of future empty triangles, or a triangle-count recurrence. Its conclusion is an actual `SimpleLowerBound`; the generic proof uses only the standard Lean logical axioms. The numerical doubling principle is BBL's. The contribution here is its completed geometric proof for these explicit hypotheses, using the following two-parameter realization.

Geometric proof and its formal decomposition. For $0 \leq i < q$ set

$$\beta_i = -\frac{\pi}{2} + (i + \frac{1}{2})\alpha, \quad b_i = \tan \beta_i.$$

Adjoin the lines

$$M_i : \quad y = \kappa g_i(\delta)(x - b_i), \quad g_i(\delta) = \frac{2b_i}{1 + b_i^2} + \frac{\delta}{b_i} = \sin(2\beta_i) + \delta \cot \beta_i. \quad (42)$$

Choose $\delta > 0$ sufficiently small and then $\kappa > 0$ sufficiently small relative to the fixed δ and the seed. This order of choices is part of the proof. BBL's published construction uses explicit scales; the qualitative two-scale statement here is established independently by the formal crossing analysis, rather than inferred from those fixed scales.

For two new lines put $t = b_i$ and $u = b_k$. Their intersection has x -coordinate independent of κ :

$$X_{ik}(\delta) = \frac{2tu(t + u)}{2tu(1 - tu) - \delta(1 + t^2)(1 + u^2)}. \quad (43)$$

When $tu \neq 1$, its limit is the reduced tangent $(t + u)/(1 - tu) = \tan(\beta_i + \beta_k)$, and the exact displacement is

$$X_{ik}(\delta) - \frac{t + u}{1 - tu} = \frac{\delta(t + u)(1 + t^2)(1 + u^2)}{[2tu(1 - tu) - \delta(1 + t^2)(1 + u^2)](1 - tu)}. \quad (44)$$

For sufficiently small positive δ its nonzero sign is that of $(t + u)tu$. If $tu = 1$, the crossing diverges along the specified exterior ray as $\delta \downarrow 0$; if $t + u = 0$, its coordinate is exactly

zero. Thus neither exceptional case is hidden inside a continuity argument with a vanishing denominator.

The reduced tangents, the old intercepts, the new half-grid intercepts, and $-\varepsilon, 0, \varepsilon$ determine a strictly ordered finite system of cuts. Equations (43)–(44) place every new–new crossing in its prescribed cut interval. Taking δ in the intersection of the finitely many resulting positive neighborhoods also makes the new slope factors nonzero and distinct. For fixed such δ , each old–new intersection tends to the corresponding old intercept as $\kappa \rightarrow 0$. Finite continuity then realizes all the prescribed strict crossing comparisons simultaneously. Small κ separates old and new slopes. Injectivity along each new crossing row excludes new triple intersections; old simplicity excludes the remaining ones. This proves actual simplicity and crossing order.

The count can now be separated from trigonometry. Treat the q new lines and Y_0 as $q + 1$ auxiliary supports. The integer row model places old support j at key $4j + 2$. A pair of auxiliary supports with crossing key z yields a mixed triangle with old support j when

$$|z - (4j + 2)| < 4. \quad (45)$$

Every auxiliary pair has two such old neighbors, except for at most q exterior pairs, each of which has one. Hence the number is at least

$$2 \binom{q+1}{2} - q = q^2. \quad (46)$$

The row inequalities make two sides consecutive on their supports. The two-side emptiness lemma proves that the corresponding triangle is an actual empty cell. The supporting triples are distinct. Thus (46) counts genuine triangles, not merely local crossing patterns.

It remains to preserve the seed's contribution. An old triangle not using Y_0 has a strict half-plane sign against Y_0 ; all new lines converge to Y_0 , so its three vertex signs persist for sufficiently small κ . The old triangles with a side on Y_0 require a different argument. Saturation forces consecutive old apices to have opposite heights. The positive central apex fixes the alternating parity. The crossing order identifies a new cap support for each noncentral distinguished triangle. On both cap endpoints, every new line has the parity-correct weak sign; strict signs against other old supports persist by continuity. This replaces each such old triangle by one empty triangle with the same two old supports. The central triangle itself is retained. The finite cap and noncap requirements can again be satisfied by one common positive κ .

Finally, these T retained or replaced triangles have at least two old nonhorizontal supports, while the q^2 mixed triangles have two auxiliary supports. The supporting triples cannot coincide. Their union is a duplicate-free witness of length at least $T + q^2$. Q.E.D.

The recurrence preserves the certified segment deficit:

$$\Delta(q, T) = q^2 - 1 - 3T, \quad \Delta(2q, T + q^2) = \Delta(q, T). \quad (47)$$

Here the equality concerns the transported lower count; if the output has additional uncounted triangles, its actual defect can be smaller. Thus doubling amplifies a compatible seed to larger orders while preserving its certified constant deficit. A smaller-defect seed or a new source of additional triangles is needed to improve that deficit.

7.2 The compatible eleven-line realization

The useful eleven-line count is 32, an existing result; the search started from the Honma-based construction displayed by Savchuk [25]. The additional work here is an explicit realization

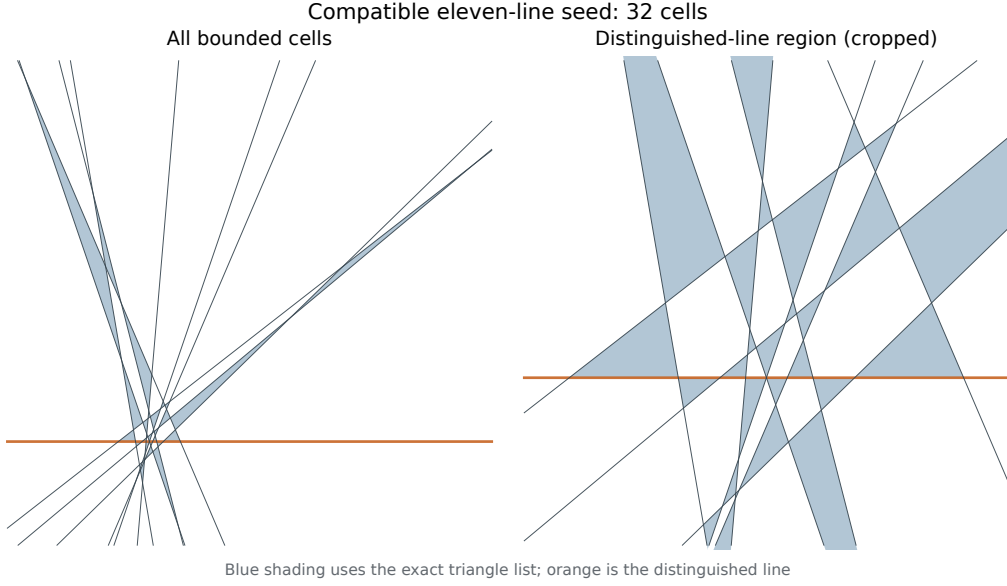


Figure 11: A finite rational realization of the compatible eleven-line seed, with global and distinguished-line views. The figure illustrates the arrangement; the all-small-parameter claim is certified by exact determinant and half-plane inequalities, not by the rendering. The count 32 is inherited from earlier work; the compatible realization and parameter certificate are the contribution recorded here.

compatible with the BBL grid for arbitrarily small positive central separation. Take $Y_0 : y = 0$ and

$$x - v_j y = a_j, \quad (v_0, \dots, v_9) = \frac{1}{10}(15, -2, 14, 1, 4, -4, 5, -3, 12, -5), \quad (48)$$

where

$$(a_0, \dots, a_9) = (-\tan \frac{4\pi}{10}, -\tan \frac{3\pi}{10}, -\tan \frac{2\pi}{10}, -\tan \frac{\pi}{10}, -\varepsilon, \varepsilon, \tan \frac{\pi}{10}, \tan \frac{2\pi}{10}, \tan \frac{3\pi}{10}, \tan \frac{4\pi}{10}). \quad (49)$$

For every $0 < \varepsilon \leq 10^{-5}$ the seed is simple, has at least 32 certified triangles, and has all nine distinguished triangles. Its central slopes are $5/2$ and $-5/2$ and its central height is $5\varepsilon/2 > 0$.

The stability check is finite but uniform in ε . The determinant for a triple of nonhorizontal lines is affine in the intercepts and therefore affine in ε :

$$D_{ijk} = (a_j - a_k)v_i + (a_k - a_i)v_j + (a_i - a_j)v_k.$$

Exact rational enclosures for the four tangent values and endpoint sign checks certify the whole interval. The first certificate used outward rational interval arithmetic; the subsequent Lean theorem uses proved trigonometric enclosures and the parametric certificate soundness theorem. The limit $\varepsilon = 0$ is used only to bound expressions; it is not presented as a simple arrangement. Five visible pairs for $w = (10, -13)$ are also proved uniformly, giving the actual exterior bound $12 : 37$ by Theorem 6.2.

Using the published BBL construction, with its finite-depth choice of the exceptional parameter, gives the ordinary mathematical family

$$q_t = 10 \cdot 2^t, \quad N_t = q_t + 1, \quad K_s(N_t) \geq T_t = \frac{100 \cdot 4^t - 4}{3}. \quad (50)$$

The arithmetic is

$$T_0 = 32, \quad T_{t+1} = T_t + q_t^2, \quad T_t = 32 + 100 \sum_{j < t} 4^j = \frac{q_t^2 - 4}{3}.$$

The BBL geometric theorem and its iteration mechanism retain their original attribution. The formula is one below $\lfloor N_t(N_t - 2)/3 \rfloor = (q_t^2 - 1)/3$; at $N_0 = 11$ the sharper known upper bound is already 32 [25]. The checkpoints 21 : 132 and 41 : 532 are weaker than the known 21 : 133 and 41 : 533. Thus the formula is not a record assertion at every member.

The quantifiers matter. For any prescribed finite depth t , one can choose the starting exceptional parameter small enough for every step up to t ; the recorded application uses $0 < \varepsilon \leq \min\{10^{-5}, (40 \cdot 2^t)^{-1}\}$. A single fixed positive ε working through infinitely many steps is neither required nor claimed. Equation (50) is supported by the verified seed and the published BBL theorem. It is not yet an end-to-end infinite-family theorem in the Lean library: Theorem 7.2 starts at $q \geq 20$, and its recursive witness interface remains to be assembled as described below.

7.3 Even companions and the stronger boundary invariant

The arrangement in (50) has defect three. Theorem 6.4 therefore gives the paper-level infinite bound

$$K_s(q+2) \geq \frac{q^2 - 4}{3} + \frac{q}{2} - 1, \quad q = 10 \cdot 2^t. \quad (51)$$

Indeed, $g(q+1, (q^2 - 4)/3) = \lceil q(q-2)/(2(q+1)) \rceil = q/2 - 1$. The stronger value observed in all completed finite examples is

$$\frac{q^2 - 4}{3} + \frac{q}{2}. \quad (52)$$

For $M = q + 2$, equations (51) and (52) equal respectively $M(M - 5/2)/3 - 2$ and $M(M - 5/2)/3 - 1$. The polynomial is the relevant Blanc upper bound for *simple* arrangements in these residue classes [2]; it is not an upper bound asserted here for unrestricted $K(M)$, and small orders can have sharper bounds.

Proposition 7.3 (Conditional full-gain even family). *If the compatible odd family admits at least $q/2$ visible pairs from an admissible normal at each stage, then*

$$K_s(q+2) \geq \frac{q^2 - 4}{3} + \frac{q}{2} \quad (q = 10 \cdot 2^t).$$

Proof. Apply Theorem 6.2 at each specified odd order.

Q.E.D.

The working boundary argument supplies a concrete route to the premise, rather than treating it as an unexplained numerical assumption. Keep $w = (10, -13)$ and require the rightmost old intercept line R to have slope $0 < m_R < 10/13$, with its rightmost vertex on Y_0 . Old visible pairs avoiding Y_0 persist under a sufficiently small pencil, and at most one visible old pair uses Y_0 . For $q = 4r$ the proposed new pairs are r negative complementary pairs, $r - 1$ positive almost-complementary pairs, the pair formed by Y_0 and the rightmost new support, and a pair formed by R and the new support at $\beta_* = \pi/4 - \alpha/2$. Their total is $q/2 + 1$, giving a net gain of at least $q/2$ visible pairs after losing at most one.

The delicate last pair is governed by the strict maximum

$$\cot \alpha \sin(2\beta) + \cos(2\beta) - 1 = \frac{\sin(2\beta + \alpha)}{\sin \alpha} - 1.$$

On the half-grid it is uniquely maximal at β_* , with gap at least $2 \sin \alpha$ from every other index. Its persistence under the δ perturbation and then the scale ratio κ/m_R is proved for actual intersections. The remaining assembly must compare this maximum with all old vertices, transfer the x -order to w -order, and preserve the rightmost-line hypothesis under

q	Odd order	Odd count	Even order	Even count	Verification
10	11	32	12	37	Lean finite witnesses
20	21	132	22	142	Lean finite witnesses
40	41	532	42	552	Lean finite witnesses
80	81	2132	82	2172	Lean finite witnesses
160	161	8532	162	8612	Lean finite witnesses
320	321	34132	322	34292	Two exact counters
640	641	136532	642	136852	See text: 642 partial

Table 6: Explicit hybrid checkpoints. The even column uses the stronger finite gain $q/2$; the infinite defect guarantee is one less. These counts describe this family, and need not be the strongest known count at the displayed order. In the last row only 641 has completed both independent exact counters.

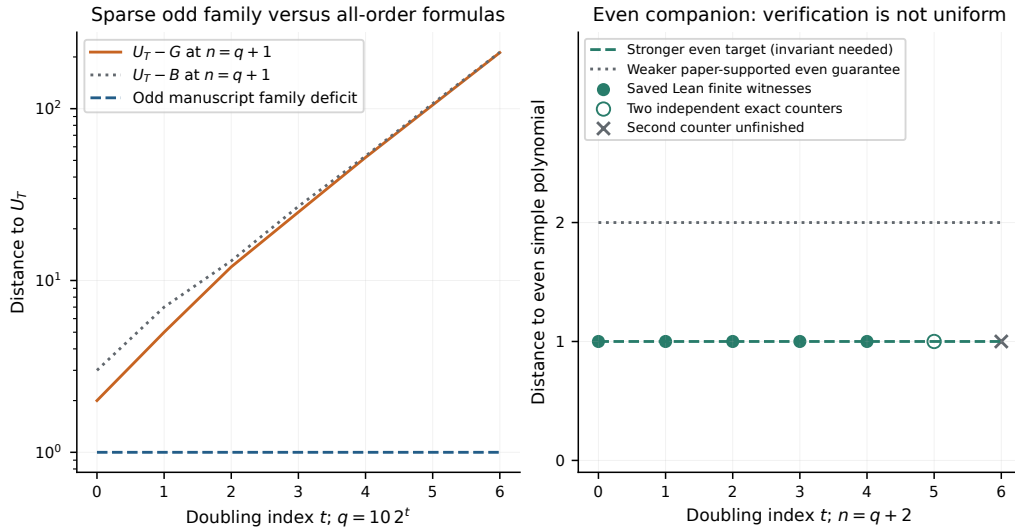


Figure 12: The sparse hybrid family compared with the all-order baselines. Odd counts are one below the classical segment bound; the stronger even checkpoints are one below the displayed simple polynomial bound. Finite Lean certificates, larger exact coordinate checks, and family formulas have different verification scopes; the plot does not turn an incomplete recursive formalization into a proved Lean theorem.

reindexing. A detailed ordinary working proof is archived, but the completed components are not presented here as an already assembled infinite Lean invariant. Proposition 7.3 makes that status explicit.

For 321, 322, and 641 lines the adjacency counter and an independently implemented direct half-plane-sign counter agree on 34132, 34292, and 136532 triangles, respectively, and simplicity is checked exactly. The largest rational components have 916, 985, and 1103 decimal digits. These are archived exact coordinate witnesses, outside the routine finite Lean catalogue. The 642-line coordinate file passed the adjacency and simplicity checks, but its independent direct count was interrupted. Its displayed value 136852 is therefore partial computational evidence, not a completed two-counter or Lean certificate. No numerical-priority claim is inferred from these larger files.

7.4 Known optimal families and independent reproduction

The classical dyadic five-line family must be kept separate from the new compatible eleven-line realization. A convenient normalized seed uses intercepts $(-1, -\varepsilon, \varepsilon, 1)$ and

$$x - v_j y = a_j, \quad (v_0, v_1, v_2, v_3) = (-15, 10, -10, 30),$$

together with Y_0 . The complete interval $0 < \varepsilon \leq 10^{-5}$ has five certified triangles, all three distinguished triangles, and two visible pairs. Its central slopes are $1/10$ and $-1/10$; the rightmost slope is $1/30$. The normalization is deliberate: an earlier affine-equivalent numerical realization had both central slopes negative, which was inadequate as an unchecked premise for the sign convention in the published iterative proof. The corrected whole-parameter seed is verified in `TamuraSeed5`.

The known optimal odd family and its simple even companions have

$$q = 4 \cdot 2^t, \quad K_s(q+1) \geq \frac{q^2 - 1}{3}, \quad K_s(q+2) \geq \frac{q^2 - 1}{3} + \frac{q}{2}. \quad (53)$$

The current finite certificates include $5 : 5$, $6 : 7$, $9 : 21$, $10 : 25$, $17 : 85$, $18 : 93$, $33 : 341$, $34 : 357$, $65 : 1365$, $66 : 1397$, $129 : 5461$, and $130 : 5525$. These reproduce known numerical consequences of the affine family of Forge and Ramírez Alfonsín and BBL’s odd-to-even discussion [1, 8]; they are not claimed as newly discovered bounds. Theorem 7.2 as currently formalized does not cover the small bases $q = 4, 8, 16$, and the unfinished large parametric 33-line draft is not counted as a completed theorem.

Parpalak and Utkin supply a different straight-line family at $q = 18 \cdot 2^t$ [18]. Its odd count is

$$K_s(q+1) \geq \frac{q^2}{3} - 1, \quad (54)$$

including $19 : 107$ and $37 : 431$. The present work independently checked their compatible 19-line seed on $0 < \varepsilon \leq 10^{-3}$ with 1632 exact endpoint determinant checks, a rational 107-triangle witness, and all 17 distinguished triangles. The interval reproduction is separate from an end-to-end Lean proof of their infinite construction. Both the seed and the family retain Parpalak–Utkin attribution. The earlier BBL statement at $18 \cdot 2^t + 1$ concerned pseudolines; it cannot be used as a straight-line result without a realization argument. Applying Corollary 6.6 to the published straight-line family gives the attributed simple even corollary

$$K_s(q+2) \geq \frac{q^2}{3} + \frac{q}{2} - 1,$$

with $20 : 116$, $38 : 449$, $74 : 1763$, and $146 : 6983$. Stronger nonsimple published values at some of these orders concern K , and do not invalidate this simple-arrangement distinction.

7.5 The exact remaining recursive formalization

The verified 21-line base has the true tangent intercepts (40) with $q = 20$, fixed rational slopes, 132 triangles, 19 distinguished triangles, ten visible pairs, and the required rightmost-line property for every $0 < \varepsilon \leq 10^{-5}$. Its finite interval checks use native evaluation; their analytic and geometric soundness proofs are separately proved. This is stronger than a single rational 21-line example.

Nevertheless, a lower-bound existential statement forgets data needed by the next step. The theorem in the library cannot be iterated merely by substituting its numerical conclusion back into its hypotheses. The remaining integration has a precise, limited form:

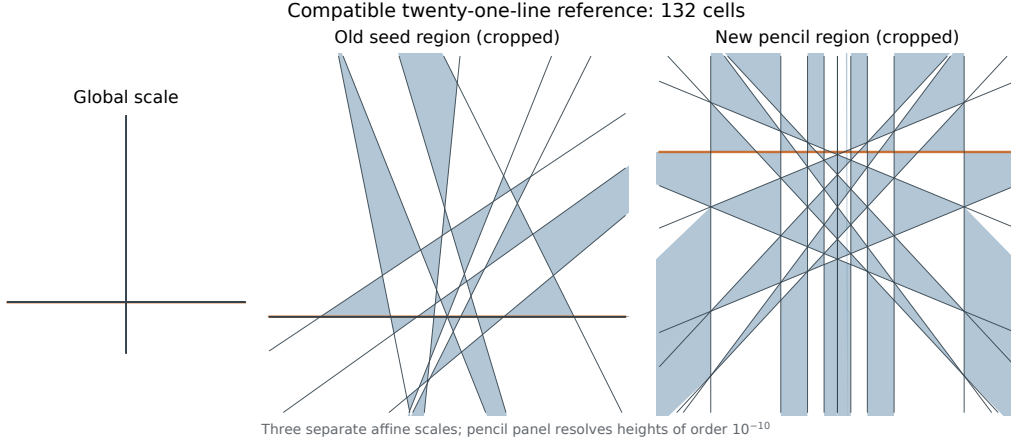


Figure 13: A 21-line hybrid checkpoint with 132 triangles. The finite global, core, and thin-pencil views illustrate the scale separation. The finite rendering is distinct from the true-tangent-grid parametric theorem in `BBLSeed21`, which establishes 132 triangles and all 19 distinguished triangles throughout its positive parameter interval.

1. Retain the constructed slopes, the duplicate-free triangle list, simplicity, distinguished-line saturation, and positive central height in a recursive seed object. Select κ to preserve the central triangle independently of the counted list, since that list may omit it.
2. Sort the old and new intercepts into the next grid. The explicit permutation and tangent-cut identity are proved in `BBLGridReindex`. The next distinguished triangles are proved in `BBLNextSaturation`: each noncentral interval is one of the mixed triangles, and the central interval uses the retained central triangle. Whole-list transport must still supply all support permutations, preserve distinctness and length, and identify the reindexed graph equations.
3. Normalize the grouped labels of `BBLSeed21` into increasing intercept order. Its central source labels are 5 and 6, whose apex has height $5\varepsilon/2$, so positivity follows without another interval search. The sorting permutation leaves the central old supports in place geometrically; hence the next central apex is exactly the old one, not merely close to it.
4. Induct on a uniform small-parameter existence statement

$$\exists \eta > 0 \quad \forall \varepsilon \in (0, \eta) \quad \exists \text{ a compatible counted seed at } (q, \varepsilon). \quad (55)$$

At the next stage shrink η below the needed half-grid gap. The slopes, δ , and κ may depend on ε .

The sought closure is, schematically,

$$\text{Compatible}(r, \varepsilon, T) \implies \text{Compatible}(2r, \varepsilon, T + (4r)^2)$$

under $r \geq 5$ and (39). This is a proposed theorem interface, not a declaration already present in the checked library. Starting from 21 lines would give the formal target

$$q_t = 20 \cdot 2^t, \quad K_s(q_t + 1) \geq \frac{q_t^2 - 4}{3}. \quad (56)$$

Numerically this is the tail of (50). Its purpose is to close the formal geometric dependency, not to establish another numerical priority claim. The stronger even family further requires the separate visibility invariant of Proposition 7.3.

A bounded search for a compatible perfect 21-line seed did not improve 132: it tested 11547910 floating proposals, with exact checking reserved for any prospective 133-triangle result, and found none. A separate 3234-program linear-feasibility transfer search likewise found no strict feasible margin. These are reproducible negative search results, consistent with the compatibility difficulties discussed in [18]; neither proves nonexistence. Improving a compatible seed, finding a useful partial-doubling interpolation, and closing the recursive invariant remain different research problems.

8 Multiplicity-sensitive upper-bound arguments

This section develops restricted upper estimates by separating the ordinary part of an arrangement from its finite multiple points. The geometric derivations below are ordinary mathematical proofs. The local fan geometry and cyclic counting are formalized in Lean, while the extraction of elementary segments from an arbitrary arrangement and the global clean-line charging argument are not. Accordingly, this section does not assert a completed Lean upper theorem for all arrangements, and none of its restricted conclusions is an improved upper bound for the full function K .

The distinction is necessary because the even-order estimates of Bartholdi–Blanc–Loisel and Blanc require simplicity [1, 2]. The classical Clément–Bader envelope is treated as a reported external theorem [3]; the local example in Section 8.6 explains why we do not silently replace its proof by a bound on all shared edges incident to a multiple point.

8.1 An exact elementary-segment identity

Definition 8.1. For an arrangement of n distinct affine lines, let t_r be the number of finite intersection points incident to exactly r lines, and let P be the number of unordered parallel pairs. A bounded *elementary segment* is the portion of an arrangement line between two consecutive finite intersection points. Let E count these segments, U count those incident to no triangular cell, and D count those incident to two triangular cells. Let a be the number of lines containing at least one finite intersection point, and put

$$S = \sum_{r \geq 3} r(r-2)t_r, \quad \delta = n(n-2) - 3T,$$

where T is the number of bounded triangular cells.

Proposition 8.2 (Segment and defect identities). *For every such arrangement,*

$$E = n(n-1) - 2P - S - a, \tag{57}$$

$$3T = E - U + D, \tag{58}$$

$$\delta = 2P + S + a - n + U - D. \tag{59}$$

If every line meets another line, then $a = n$ and $\delta = 2P + S + U - D$.

Proof. Each nonparallel unordered pair contributes to exactly one finite point, so

$$n(n-1) - 2P = \sum_{r \geq 2} r(r-1)t_r.$$

If a line has $s > 0$ distinct finite intersection points, it contributes $s - 1$ bounded elementary segments; an isolated line contributes none. Summing point-line incidences gives

$$E = \sum_{r \geq 2} rt_r - a.$$

Subtracting these expressions yields the first identity. Each triangular cell uses three elementary segments. A segment is incident to zero, one, or two such cells, so the total number of triangle-side incidences is $E - U + D$. Substitution into the definition of δ gives the last identity. Q.E.D.

The term D cannot be dropped in a nonsimple arrangement. Multiple points reduce the available segment count by S , but sharing can compensate for some of that loss. This is precisely the interaction that a classical upper proof must control.

Lemma 8.3 (A shared side has a multiple endpoint). *An elementary segment incident to two bounded triangular cells cannot have both endpoints ordinary, where an ordinary point is incident to exactly two arrangement lines.*

Proof. At each ordinary endpoint there is a unique supporting line other than the line containing the segment. Both adjacent triangles would therefore have the same three supporting lines. Three distinct lines have at most one bounded triangular region, whereas the two incident cells lie on opposite sides of the shared segment. This is impossible. Q.E.D.

For the remainder of the section assume that $n \geq 4$ is even and that all pairs of lines are nonparallel. Let the *core* consist of the finite points of multiplicity at least three. Write

$$q = \sum_{r \geq 3} t_r, \quad I = \sum_{r \geq 3} r t_r,$$

and let h count the lines meeting the core. Let D_1 count shared elementary segments with exactly one core endpoint and D_2 those with two core endpoints. By Lemma 8.3, $D = D_1 + D_2$, and hence

$$\delta = S + U - D_1 - D_2. \quad (60)$$

8.2 Parity on a line outside the core

A line is *clean* if it contains no core point. The following argument adapts the local parity method in Blanc's Section 2 [2]. Its proof is included because the transverse lines may have multiple intersections away from the clean line.

Lemma 8.4 (Clean-line charging). *For even $n \geq 4$ and pairwise nonparallel lines,*

$$n - h \leq 2U + D_1. \quad (61)$$

Proof. First fix a clean line L , place it horizontally, and number its $n - 1$ crossings from left to right. Let R_i be the transverse line at crossing i . Denote the bounded segment of L between crossings i and $i + 1$ by ℓ_i , and its two exterior rays by ℓ_0 and ℓ_{n-1} . Write p_i^+ and p_i^- for the faces above and below ℓ_i , and r_i^+, r_i^- for the elementary transverse pieces of R_i immediately above and below L . A transverse piece may be a bounded segment or an unbounded ray.

We show that some bounded $r_i^\pm$ is unused or shared. Suppose otherwise: every bounded transverse piece is incident to exactly one triangle. Each bounded ℓ_i is unshared by Lemma 8.3.

If both p_1^+ and p_1^- are nontriangular, then r_1^+ and r_1^- are unused: their other adjacent faces meet the exterior ray ℓ_0 . At least one of these transverse pieces is bounded, because R_1 meets another transverse line. This contradicts the supposition. Reflecting across L if necessary, assume p_1^+ is triangular; p_1^- is then not.

For $1 \leq i \leq n - 2$, prove the following assertions together:

- (i) if i is even, p_i^+ is nontriangular; if i is odd, p_i^- is nontriangular;

- (ii) if both faces at ℓ_{i-1} are nontriangular, then p_i^- is triangular for even i , and p_i^+ is triangular for odd i .

The base case follows from the choice of p_1^+ and the unbounded faces at ℓ_0 . Consider an even step; the odd step exchanges signs. If p_{i-1}^+ is triangular, then p_i^+ cannot be triangular, since otherwise the two triangles share the bounded piece r_i^+ , contrary to the supposition. Assertion (ii) has a false antecedent in this case.

Otherwise both p_{i-1}^+ and p_{i-1}^- are nontriangular by the preceding odd step. This cannot happen for $i = 2$, so $i \geq 4$. The earlier even step makes p_{i-2}^+ nontriangular. Hence r_{i-1}^+ is unused, and must be unbounded. The intersection of R_{i-1} and R_i must consequently lie below L . It follows that r_i^- is bounded. Its left adjacent face is nontriangular, so its right adjacent face p_i^- must be triangular. Since ℓ_i is unshared, p_i^+ is nontriangular. This proves both assertions.

Now $n - 2$ is even, so p_{n-2}^+ is nontriangular. If p_{n-2}^- is also nontriangular, both pieces at the last crossing are unused and at least one is bounded, again a contradiction. Otherwise p_1^+ and p_{n-2}^- are triangular, while r_1^- and r_{n-1}^+ are unused. The lines R_1 and R_{n-1} intersect. If their intersection is below L , then r_1^- is bounded; if above, then r_{n-1}^+ is bounded. This is the final contradiction.

Charge each clean line to one bounded unused or shared transverse segment supplied by this argument. An unused segment can receive at most two charges, one at each ordinary endpoint. A shared segment must have a multiple endpoint, so it can receive at most one charge if counted by D_1 , and none if counted by D_2 . There are $n - h$ clean lines, giving (61). Q.E.D.

Remark 8.5 (Why parallel pairs have been excluded). Three parallel vertical lines and one horizontal line give a clean horizontal line whose transverse pieces are all unbounded. Thus simply removing parallel-participating lines from the set to be charged does not prove the lemma. The identities of Proposition 8.2 allow parallels, but the charging argument above does not. Every subsequent upper estimate in this section retains the pairwise nonparallel hypothesis.

Combining (60) and (61) gives the basic budget

$$2\delta \geq n + 2S - h - 3D_1 - 2D_2. \quad (62)$$

The remaining problem is to bound the shared-side contribution at the core.

8.3 The geometric obstruction in a cyclic fan

At an r -fold point v , the incident lines determine $2r$ rays in cyclic order, with opposite rays separated by r positions. Let $d_1(v)$ count shared elementary segments starting at v and ending at an ordinary point, and let $d_2(v)$ count those ending at another core point. A ray supports at most one such incident elementary segment.

Lemma 8.6 (No long run of ordinary-ended shared rays). *For $r \geq 3$, no $r - 1$ consecutive rays at an r -fold point can all support segments counted by $d_1(v)$.*

Proof. Such a run would border r consecutive triangular sectors. At the ordinary endpoint of each selected shared segment, the opposite supporting lines of the two neighboring triangles must coincide: besides the radial line through v , only one arrangement line passes through that endpoint. Propagating this equality makes the opposite sides of all r triangles lie on one line M .

The line M would meet $r + 1$ consecutive rays from v , which include an opposite pair. A line meeting both open opposite rays must contain their connecting segment and hence v . But the opposite side of a nondegenerate triangle with vertex v cannot lie on a line through v . This contradiction proves the lemma. Q.E.D.

This is an actual geometric statement, not an assumed combinatorial prohibition. The Lean module `FanGeometry` proves uniqueness and propagation of opposite supports for real lines and points. The module `CyclicFan` supplies the cyclic-to-geometric bridge from triangular sectors, antipodal rays and ordinary selected endpoints.

Proposition 8.7 (Local fan budget). *At every finite point of multiplicity $r \geq 3$,*

$$d_1(v) \leq 2r - 3, \quad 3d_1(v) + d_2(v) \leq 6r - 6. \quad (63)$$

Proof. Count selected rays in all $2r$ cyclic windows of length $r - 1$. Each window contains at most $r - 2$ selected rays by Lemma 8.6, and each selected ray belongs to exactly $r - 1$ windows. Therefore

$$(r - 1)d_1(v) \leq 2r(r - 2).$$

If $d_1(v) \geq 2r - 2$, the left side is at least $2(r - 1)^2 = 2r(r - 2) + 2$, a contradiction. Finally $d_1(v) + d_2(v) \leq 2r$, so

$$3d_1(v) + d_2(v) = 2d_1(v) + (d_1(v) + d_2(v)) \leq 6r - 6.$$

Q.E.D.

The cyclic double count is proved in `FanCount`. The theorem `CyclicFan.selected_card_1e` combines it with the real geometric obstruction, without assuming the no-long-run conclusion as an unexplained premise.

8.4 A weighted bound for an arbitrary finite core

Proposition 8.8 (Multiplicity-weighted defect estimate; manuscript geometry). *Let $n \geq 4$ be even and let the arrangement be pairwise nonparallel. Then*

$$2\delta \geq n + \sum_{r \geq 3} (2r^2 - 11r + 6)t_r. \quad (64)$$

Proof. Each D_1 segment has exactly one core endpoint and each D_2 segment has two. Summing (63) gives

$$3D_1 + 2D_2 \leq 6I - 6q.$$

Also $h \leq I$, since every core line contributes at least one point-line incidence. Inserting both inequalities into (62) yields

$$2\delta \geq n + 2S - 7I + 6q,$$

which is (64) after expanding S, I, q .

Q.E.D.

The local weights are $-9, -6, 1, 12, 27$ at multiplicities $3, 4, 5, 6, 7$, respectively. Consequently the obstruction to recovering a simple-style parity defect is concentrated at triple and quadruple points; higher multiplicity has a positive weight in this accounting.

Corollary 8.9 (Nonnegative core weight; manuscript geometry). *Under the hypotheses of Proposition 8.8, if*

$$\sum_{r \geq 3} (2r^2 - 11r + 6)t_r \geq 0,$$

then

$$T \leq \left\lfloor \frac{n(n-5/2)}{3} \right\rfloor.$$

In particular, this holds if every finite multiple point has multiplicity at least five.

Proof. The weighted estimate implies $\delta \geq n/2$, giving the displayed upper bound by the definition of δ . For $r \geq 5$, the exact factorization

$$2r^2 - 11r + 6 = 1 + (r-5)(2r-1) \geq 1$$

proves the stated sufficient condition. Q.E.D.

This extends the same polynomial to a restricted nonsimple class. It says nothing comparable for a general core consisting of many triple points. Numerical first priority for this restricted consequence has not been established.

8.5 A sharper estimate with at most two core points

Lemma 8.10 (A fan with at most one core-to-core ray). *If $d_2(v) \leq 1$ at a point of multiplicity $r \geq 3$, then $d_1(v) \leq 2r - 4$.*

Proof. Mark which of the $2r$ sectors at v are triangular. A ray supports a shared elementary segment exactly when both neighboring sectors are triangular. If every sector is triangular, every ray is shared; deleting at most one d_2 ray leaves a run of at least $2r - 1$ selected rays, contradicting Lemma 8.6. Hence a nontriangular sector exists.

Suppose $d_1(v) \geq 2r - 3$. Two separated nontriangular sectors exclude at least four shared rays, as do three consecutive nontriangular sectors. Both situations are impossible. If exactly one sector is nontriangular, the shared rays form a path of length $2r - 2$. Deleting at most one d_2 ray leaves at most two selected runs. Their total capacity is $2(r - 2)$, less than the required $2r - 3$. If exactly two adjacent sectors are nontriangular, at most $2r - 3$ shared rays remain. All would have to be selected, forming a run of length $2r - 3 > r - 2$. These cases exhaust the possibilities and prove the lemma. Q.E.D.

Theorem 8.11 (At most two finite multiple points; manuscript geometry). *An arrangement of an even number $n \geq 4$ of pairwise nonparallel affine lines with at most two finite multiple points satisfies*

$$T \leq \left\lfloor \frac{n(n-5/2)}{3} \right\rfloor + 1. \tag{65}$$

More precisely, the defect bounds are $\delta \geq n/2$ if $q = 0$, $\delta \geq n/2 - 1$ if $q = 1$, and $\delta \geq n/2 - 3$ if $q \leq 2$.

Proof. With at most two core points, at most one ray at a core point can terminate at another core point. Lemma 8.10 gives $D_1 \leq 2I - 4q$. There is at most one core-to-core shared segment, so $D_2 \leq q(q-1)/2 \leq 1$. Moreover,

$$h \leq I - D_2,$$

because any such segment lies on a line counted twice in the core-incidence total and once among the core lines. Substitution in (62) yields

$$2\delta \geq n + \sum_{r \geq 3} (2r^2 - 11r + 12)t_r - D_2.$$

For $r \geq 3$,

$$2r^2 - 11r + 12 = -3 + (r - 3)(2r - 5) \geq -3.$$

Thus $2\delta \geq n - 3q - D_2 \geq n - 7$. Since n and 2δ are even, $2\delta \geq n - 6$. Rearranging gives (65). If $q = 1$, then $D_2 = 0$ and the inequality $2\delta \geq n - 3$ rounds to $2\delta \geq n - 2$; if $q = 0$, it gives $2\delta \geq n$ directly. Q.E.D.

The hypotheses are attained by externally attributed exact constructions. Table 7 records examples whose pairwise nonparallel status, triangle count, and two triple points were checked exactly. Their equality with (65) proves sharpness within this restricted class, not optimality over all classical arrangements.

n	T	Restricted upper	Coordinate provenance
8	15	15	Earlier construction, retained in the current gallery
14	54	54	Maiorana, two-triple-point witnesses
26	204	204	Parpalak–Utkin gallery
50	792	792	Parpalak–Utkin gallery

Table 7: Attainment in the class covered by Theorem 8.11. Sources: Maiorana [16], Parpalak and Utkin [20]. These are not new numerical records claimed by this paper.

ProofAtlas reports a stronger restriction-specific result at $n = 14$, covering up to four finite multiple points [21]. Its proof was not independently reproduced here. Our two-point estimate covers all even orders and arbitrary core multiplicities, but neither that fact nor the table establishes priority for the restricted theorem.

8.6 A checked shared-fan example and the limits of smoothing

Consider the six lines

$$x = 0, \quad y = 0, \quad x + y = 3, \quad x - y = 0, \quad 2x + y = 3, \quad x + 2y = 3. \quad (66)$$

They are the three sides of a triangle and its three medians. There are six bounded triangular cells and four triple points. At $v = (1, 1)$, six distinct nonzero elementary radial segments are shared sides, each belonging to two of those cells. Three have ordinary other endpoints and three have multiple other endpoints, giving $d_1(v) = d_2(v) = 3$; globally $D_1 = D_2 = 3$.

The **SharedFan** module proves the real geometric facts: each designated triangle has nonempty uncut interior, the six interiors are pairwise disjoint, the radial segments are distinct and elementary, and each is a side of two distinct triangles. The proof uses ordinary kernel checking.

This matters when interpreting the Clément–Bader proof. Its local shared-pair sentence cannot literally bound all shared segments incident to a multiple point by two. An intended assignment of each shared side to one endpoint would be a different assertion requiring global justification. The example has only six triangles, below the reported seven-triangle upper bound at order six; it is therefore not a counterexample to that upper theorem. The present paper leaves that distinction explicit and uses the separate D_1, D_2 accounting instead.

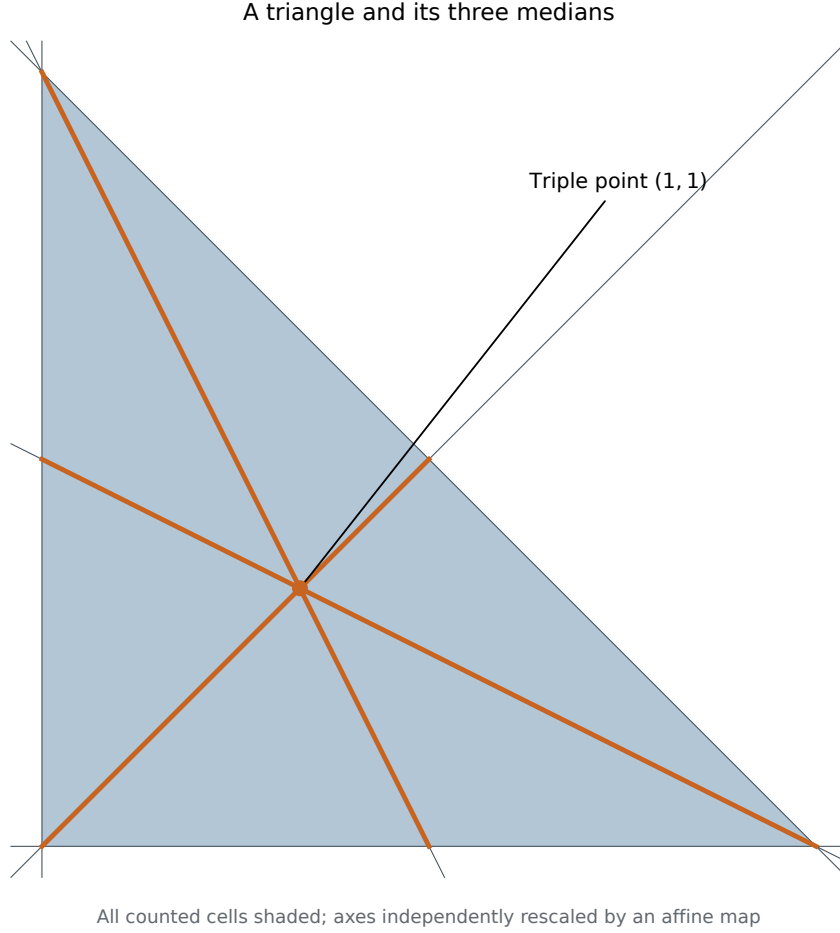


Figure 14: The exact arrangement (66). Six triangular cells meet around the central triple point, and all six radial sides are shared. This refutes only a literal local incidence reading of the shared-pair statement in Clément–Bader’s Lemma 1; it does not refute their final upper theorem. The figure is newly generated from the integer equations.

Nor can one justify a simple upper bound merely by perturbing every multiple point. For a retained two-triple-point Maiorana 14:54 witness, the four local generic resolution types have triangle counts 52, 52, 53, 52. These counts were checked with exact arithmetic; they are finite experimental evidence, not a new Lean theorem classifying all perturbations. They demonstrate why a universally nondecreasing smoothing argument needs more than the existence of a small generic perturbation. Our global upper argument never assumes such monotonicity.

8.7 Formal status and the remaining bridge

The following division is part of the mathematical claim, not an implementation footnote.

In particular, `CleanLineBudget` also contains arithmetic implications with a symbolic parallel-pair term. Those implications do not prove the missing charging lemma with parallels. The nine conditional arithmetic theorems and the geometric fan modules use only the standard axiom set recorded for this development. Exact finite checks on 597 test arrangements, followed by the imported Maiorana and gallery witnesses, support the accounting and expose implementation mistakes; they do not replace its general proof.

The main unresolved direction is therefore precise. One must control the negatively weighted triple and quadruple points, account for the entire core rather than a chosen sub-

Module or argument	Completed	Still required for a global Lean upper theorem
SharedFan	Exact real six-line counterexample to the literal local reading.	No general conclusion is inferred from this example.
FanGeometry	Uniqueness and propagation of opposite supports; obstruction at opposite rays.	Extract fan strips from arbitrary arrangement cells.
FanCount	Cyclic-window double count and $2r - 3$ bound.	Supply its geometric selected set.
CyclicFan	Actual sector geometry implies the no-long-run property and selected-cardinality bound.	Construct the cyclic sector data globally.
CleanLineBudget	Integer consequences of explicitly supplied incidence, charging, and fan inequalities.	Prove those hypotheses for all arrangements in the intended class.
Paper-level geometry	Segment identity, clean-line parity charging, and restricted consequences above.	Formal elementary edges, their cell incidences, global summation, and the sharper $d_2 \leq 1$ fan refinement.

Table 8: The upper-bound formalization boundary. Every completed module in this table was checked without placeholders; the table does not identify the conditional arithmetic module with a completed geometric upper theorem.

graph, and separately handle parallel classes. Completing the global Lean extraction would certify the restricted results above. Improving the unrestricted numerical upper envelope requires an additional mathematical idea beyond those formalization tasks.

9 Exact certificates, external reproduction, and unsuccessful searches

The experimental program serves two purposes: producing concrete coordinate witnesses for lower bounds, and identifying which proposed general arguments survive exact checks. These purposes require different standards of evidence. A successful coordinate witness proves an existential lower bound at its specified order. A failed search ordinarily proves neither an upper bound nor the nonexistence of a better construction. This section records both positive and negative outcomes without merging their logical scopes.

9.1 From a numerical proposal to an exact geometric witness

The stored coordinates use the convention $a_ix + b_iy = c_i$, with rational or integer coefficients. Each line is normalized to a primitive integer triple. Pair intersections are computed as homogeneous integer triples, normalized with a positive final coordinate. Consequently, collinearity, concurrence, parallelism, and comparisons along a supporting line can all be tested without floating-point tolerances.

Two independently implemented counters are used. The first sorts the distinct intersection vertices on each line and constructs the graph of elementary bounded segments; a three-cycle supported by three different lines is a triangular cell. The second enumerates support triples and tests whether every other line has a common weak sign at their three vertices. Nonzero area and the distinction between open interiors and boundaries are checked explicitly. Where a complete independent verification is reported, both counters agree on the triangle set, not merely on a rounded area or a visual estimate. The Lean geometric soundness theorem

provides a third, logically separate link from accepted coordinate predicates to real affine triangles.

Numerical optimization is used only to propose coordinates or rank candidate chambers. A proposal is rationalized and recounted before promotion. In the phase/chart experiments, progressively finer integer scales were accepted only when the entire exact triangle set and simplicity were retained. Such a finite acceptance does not prove that an arbitrary rounding of an irrational construction is valid. The real all-order theorem for G has its own determinant, trigonometric, and counting proof.

The released catalog contains 138 distinct promoted coordinate identities. Several identities can occur at the same order and triangle count, and an identity may have several repository source paths. Thus 138 is a catalog size, not a count of new records or independent discoveries. The complete catalog in Table 11 includes simple versus nonsimple status, provenance collections, and coordinate-hash prefixes; its machine-readable companion records full hashes, source paths, and theorem modules. The finite Lean certificates use native evaluation where indicated. This trust boundary is broader than that of the universal construction, whose proof uses only Lean’s standard logical axioms.

9.2 Finite improvements relative to the retained collection

The phase and affine-chart work raised the preceding saved values at eleven orders:

n	39	47	48	51	53	54	55	59	60	99	195
previous	469	690	720	817	884	918	954	1102	1140	3169	12481
current	470	691	721	818	885	919	955	1103	1141	3170	12482

Each row is an improvement over the project’s preceding retained coordinate catalog. The underlying trigonometric family is due to Füredi–Palásti [10]; these comparisons do not establish numerical first priority. The general formula G now explains all eleven displayed counts. The first projective search reached 47 : 691 after 11,264 proposals; the subsequent exposed-cap description replaced that search by a direct construction and an all-order proof.

The earlier OEIS-attributed values 28 : 238, 30 : 275, and 34 : 357 remain independently certified [17, 28]. The drawings in this paper render the currently retained exact witnesses, obtained by exterior extension of retained odd-order inputs. They are not asserted to be literal reproductions of the March diagrams. Likewise, the historical inventory in Table 12 preserves earlier outputs without upgrading every original derivation to a completed general theorem.

The current strongest saved simple and classical coordinate counts for every $3 \leq n \leq 60$ appear in Table 10. An absent entry in the explicit OEIS table is recorded as a dash. Absence from that table is not evidence that a value is absent from the literature; conversely, a coordinate certificate does not establish optimality without an upper theorem covering the same arrangement class.

9.3 Independent reproduction of external constructions

The current Parnalak–Utkin coordinate gallery [18] supplied exact witnesses at

$$\begin{aligned} &8 : 15, \quad 14 : 54, \quad 20 : 117, \quad 26 : 204, \quad 32 : 315, \\ &36 : 402, \quad 38 : 450, \quad 42 : 553, \quad 46 : 667, \quad 50 : 792. \end{aligned}$$

Both independent counters reproduced their triangle sets. These witnesses retain their source attribution; in particular, the eight-line numerical value has older discovery priority

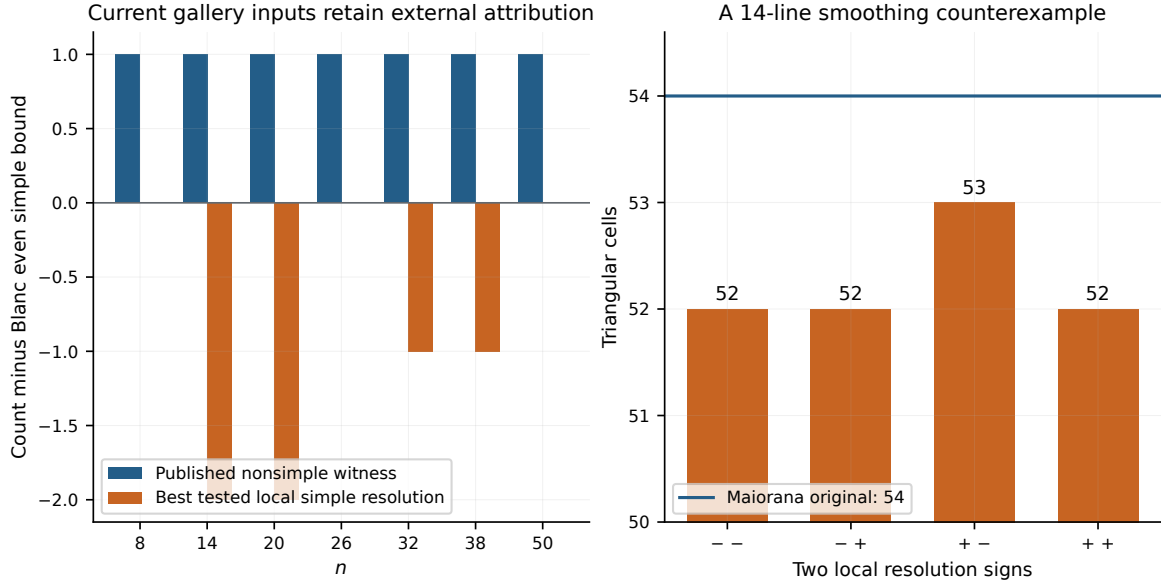


Figure 15: Exact local-resolution tests on attributed external constructions. Left: Parpalak–Utkin gallery witnesses and their best tested simple local resolutions, measured relative to Blanc’s even simple-arrangement polynomial; the eight-line value has older priority. Right: all four local resolutions of Maiorana’s two-triple-point 14 : 54 example. The original arrangements are nonsimple, so exceeding the simple polynomial is consistent. These are smoothing obstructions, not upper bounds for the classical problem.

than the coordinate gallery. Their addition to this repository improves its verified coverage, not the discoverer’s claim to the numbers. The inspected gallery snapshot is commit `feae4f5571a988a3ed45bd26a36259ebb7d7d3d2`.

All fifteen of Andrea Maiorana’s published 14 : 54 coordinate configurations [16] were also independently checked. Fourteen have exactly two triple points, and one has four; none has a parallel pair. The source snapshot is commit `e47c7cfd9661e54d29befb83118bf83d5f804028`. Its CC BY 4.0 licence, original coordinate files, and source attribution are retained. Added wrappers record counts and verification metadata. This reproduction imports no claim that a simple-arrangement upper theorem applies to these nonsimple witnesses.

These examples also expose a concrete limitation of smoothing. In Maiorana’s first example, the triple supports are $\{0, 4, 10\}$ and $\{0, 11, 12\}$. Independently shifting the offsets of lines 4 and 11 by $\pm 1/1000$ realizes all four local resolution choices while preserving every originally nonzero determinant sign. The resulting triangle counts are 52, 52, 53, 52, all below 54. Thus the proposed nondecreasing local smoothing strategy fails even when the local sign choices are independently realizable by straight lines. The argument identifying these sign choices with all sufficiently small local resolutions is an ordinary finite sign-stability argument, not a completed Lean theorem.

For the gallery’s collinear-core examples at orders 8, 14, 20, 26, 32, 38, 50, all 132 recorded local resolutions were realized by exact private-line offsets. Their respective best simple counts were 14, 51, 114, 203, 313, 448, 791. In particular, a universal claim that resolving such a core loses at most one triangle is also false. Figure 15 separates these finite obstructions from any general upper-bound assertion.

9.4 Projective charts, algebraic curves, and chamber searches

Projective geometry provides a useful global neighborhood: changing the line at infinity moves all lines together and may bound formerly unbounded triangular faces. It can also lose bounded cells. For the exact saved 81 : 2132 and 161 : 8532 hybrid arrangements, the projective triangular-face census already equals their bounded-cell count; a chart change alone cannot increase those particular counts.

Exact rational linear-arithmetic searches give sharper restrictions for some fixed configurations. The nine-line half-phase rational arrangement has 21 projective triangular faces, but the recorded QF_LRA solver result rules out a chart retaining all 21. Similar fixed-seed tests find no gain for the tested half-phase arrangements at 14, 20, 26, 44, and no chart retaining all 471 projective triangles of the saved 39 : 470 witness. For a nonuniform cubic 48-line seed with 724 projective triangles, 2,903 exact linear-arithmetic checks rule out 722 bounded triangles within the tested chart-retention formulation. These are exact-solver results about specified rational seeds. Their UNSAT certificates have not been independently checked in Lean, and they are not global upper bounds at those orders.

The algebraic-curve direction tested both nodal-cubic angle perturbations and torsion cosets on real elliptic curves. A larger projective face count did not automatically produce a better affine construction. For the better two-real-component elliptic examples, the projective, original affine, and best found affine counts were respectively

n	42	48	54	60
projective	550	724	922	1144
original affine	538	710	907	1127
best found affine	539	712	908	1129

All fall below the corresponding retained affine bounds. The chart scans at 54 and 60 had 90-second limits. Numerical ranking and bounded running time preclude a global nonexistence inference.

Oriented-matroid mutations and kinetic chamber sweeps addressed a different limitation: moving inside a fixed projective type cannot create new projective triangular faces. The mutation searches changed determinant signs, using linear-programming realization where appropriate. Kinetic sweeps varied many offsets, or both slopes and offsets, and visited concurrency events along selected rays. Their event counts are not counts of all realizable order types, and unsuccessful floating rays were discarded. Figure 16 records effort without treating heterogeneous search units as comparable coverage measures.

The final 39-line projective beam examined 2,122,270 candidate edges through 4,559 states without exceeding 470 affine or 471 projective triangles, even at its combinatorial scoring stage. The 48-line LP walk accepted 471 mutations out of 511 attempts and retained 721 affine and 724 projective triangles. At 44 lines, singular-boundary searches included 35,352 paired collapses; the retained beam checkpoint had 771 states and 446,422 candidate edges, without exceeding 608. An unfinished round was discarded. These results motivate a change of neighborhood or a structural argument rather than repeating the same local searches.

Other bounded approaches included Fourier deformations (2,687,600 proposals in one 39-line run), partial deletions from BBL constructions, and constrained tangent-grid transfer by LP/MILP. The tested optimal 23-line chart transfers did not admit a positive-margin realization on the desired grid; a constrained near-optimal search gave 144 triangles, below a separately known compatible 145. None of these observations proves nonexistence of another compatible seed or a different successful deletion pattern.

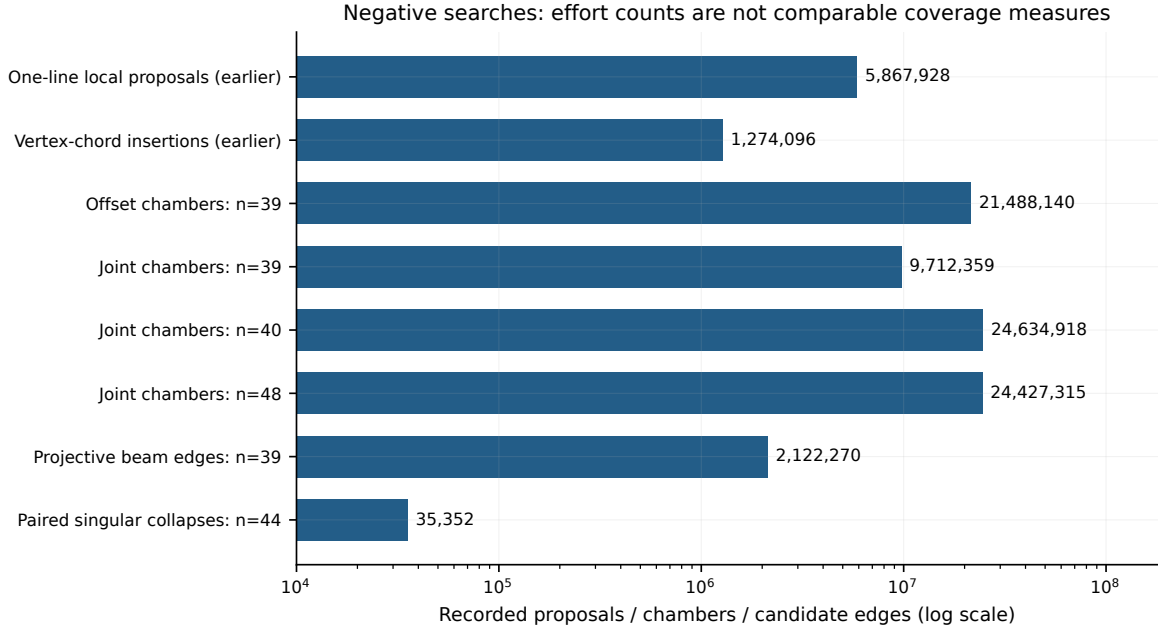


Figure 16: Selected unsuccessful searches, with their recorded units. Proposal, chamber, and candidate-edge totals describe different search neighborhoods. They do not measure exhaustive coverage, justify an upper bound, or rank the mathematical promise of the methods. All proposed improvements would require subsequent exact geometric verification.

9.5 Reproduction and presentation audit

The research registers preserve source snapshots, experiment scripts, seeds, parameters, checkpoints, and exactness limits [29]. The paper’s figures are original vector renderings, not copied source figures. The generator `paper/scripts/generate_figures.py` reads the research data without modifying it and exports PDF figures, PNG previews, LaTeX tables, and machine-readable rendering data. The figure manifest records input-file hashes and the mathematical status of each caption. Geometric views recompute their exact triangle lists before converting coordinates to floating point for display. Cropped detail panels and independent affine axis rescaling are explicitly labeled; lengths and angles are not inferred from those views.

The appropriate experimental conclusion is therefore selective. Exact coordinates have strengthened the retained lower-bound envelope and revealed specific failed construction rules. The successful universal advance comes from replacing a numerical chart search by a geometric exposure proof and an all-order count. The remaining negative searches narrow useful next steps, but they do not solve the general optimization problem.

10 Formal verification and reproducibility

10.1 The verified snapshot

The mathematical snapshot used by this paper is repository commit `99fdc8ec1ef8b1fb22c3da32b011b7361762e958`. It pins Lean 4.31.0 and Mathlib commit `fabf563a7c95a166b8d7b6efca11c8b4dc9d911f`. The local release audit completed on 22 September 2026, and the independent GitHub Actions run `35699093671` subsequently passed. These are reproducibility statements about a particular collection of declarations and coordinates, not an assertion that every conjecture discussed in this paper has been formalized.

The audit built 220 targets and recorded hashes for all 325 active Lean source files. It inspected 4,227 theorem declarations: 3,412 use only the accepted standard logical axioms, while 815 also depend on explicitly recorded native evaluation. These are counts of theorem declarations, including auxiliary and generated declarations, not counts of independent mathematical contributions. The coordinate audit checked 138 distinct coordinate identities with two exact counters; 104 of the identities also satisfy the simplicity check.

10.2 Three proof and computation boundaries

General symbolic proofs. The arbitrary-order construction, the actual geometric BBL step, and the local fan geometry use the usual Lean axioms `propext`, `Classical.choice`, and `Quot.sound`. No unproved geometric construction is installed as an axiom. Their conclusions are not inferred from a finite range of successful tests. Lean and Mathlib provide the real analysis, finite combinatorics, and algebra used in these proofs [5, 27].

Finite certificates. Large coordinate checks use `native_decide`, which trusts Lean’s native compiler and runtime in addition to the ordinary logical kernel interface. The general certificate soundness theorem is proved symbolically; native evaluation verifies the finite integer sign conditions and list properties. The audit reports these dependencies rather than describing all certificate computation as kernel reduction. Small examples and many formula comparisons also have direct kernel-reduction proofs.

Computational and paper-level evidence. Exact rational searches, interval calculations, ordinary geometric arguments, and the experimental large witnesses are retained with separate status. They are not imported as axioms into the all-order theorem. In particular, the incomplete 642-line recount and the uncompiled real-parameter 33-line draft are not active certificate results. The ordinary upper-budget derivation is distinguished from the already checked local fan lemmas.

10.3 What the audit rejects

The active-source scan strips comments and strings and rejects proof placeholders, admissions, custom axiom declarations, and unsafe definitions. The additional Lean audit recursively collects the axioms used by every theorem in the `Kobon` namespace and rejects unapproved dependencies. It permits and records the standard logical axioms and the generated native-evaluation dependencies just described.

The archived March file is deliberately outside the active library. Preserving it is evidence of the development history, not a way to conceal its eight placeholders inside an otherwise successful build. Likewise, an unfinished draft is kept under the research directory with an explicit status note rather than omitted silently from an allegedly complete theorem library.

10.4 Theorem-level correspondence

Table 9 gives the principal correspondence between the paper and the Lean development. The public repository contains more detailed logs and dependency information; the table is intended to make each major mathematical claim traceable without requiring a reader to inspect thousands of auxiliary declarations.

Table 9: Principal mathematical claims and formal sources. Names refer to the frozen mathematical snapshot.

Mathematical claim	Formal source
Exact certificate soundness	<code>Geometry.validate_sound; Simple.validate_simple_sound</code>
Certificates are nonoverlapping cells	<code>Cells.interior_eq_signCell; Cells.distinct_interiors_disjoint</code>
Classical affine construction	<code>FurediPalasti.lower_bound; FurediPalastiCount.ordered_bound</code>
Shifted repeated-index gain	<code>ShiftedCount.ordered_bound_plus; ShiftedFurediPalasti.simple_lower_bound</code>
Actual projective transformation	<code>Projective.oriented_transform; Projective.triangle_preserved</code>
One exposed cap	<code>FurediPalastiWrap.one_cap_simple_lower_bound</code>
Two exposed caps	<code>FurediPalastiTwoCaps.two_cap_simple_lower_bound</code>
Universal G and exact gain	<code>Universal.baseline_sound; Universal.baseline_improvement</code>
Finite-enhanced all-order bound	<code>Universal.all_n; AllN.dominates_every_saved_certificate</code>
Unconditional 49-seed target	<code>AllN.from_49_unconditional; Universal.dominates_49_target</code>
Exterior addition from visible pairs	<code>Exterior.extension</code>
Real compatible seed family	<code>SeedFamily; Parametric; TangentBounds</code>
Geometric one-step BBL doubling	<code>BBLDoubling.doubling</code>
Next-grid saturation component	<code>BBLNextSaturation.next_saturated</code>
Local fan geometry and budgets	<code>SharedFan; FanGeometry; FanCount; CyclicFan; CleanLineBudget</code>
Preserved historical finite bounds	<code>Results.earlier_NNN; Results.best_classical_NNN; Results.best_simple_NNN</code>

The exterior theorem assumes a valid visible-pair list, not an unproved assertion that enough visible pairs always exist. The BBL theorem assumes the stated tangent-grid and saturation conditions, not its own desired conclusion. Conversely, the upper-budget arithmetic deliberately accepts global incidence inequalities as hypotheses where the arrangement-to-budget bridge is unfinished. Inspecting hypotheses is as important as inspecting the absence of placeholders.

10.5 Reproducing proofs, coordinates, and figures

After cloning the repository and installing the pinned Lean toolchain, the principal checks are:

```
lake exe cache get
python scripts/verify_lean.py --jobs 2
python scripts/verify_coordinates.py
python scripts/verify_seed_interval.py
python scripts/evidence_manifest.py --check
```

The finite envelope can be regenerated from the certificate index using `scripts/generate_all_n.py`. This generator emits theorem references; Lean checks the resulting proposition

independently of the script’s intentions. The generated source is not a trusted oracle for geometry.

All sixteen scientific figures in the initial manuscript are generated by `paper/scripts/generate_figures.py`. The script reads the saved formulas, coordinates, and experiment records. For geometric renderings it fixes the triangle combinatorics using exact rational arithmetic before converting coordinates to floating point for plotting. Vector PDF is the publication format; PNG images are previews. Rational proxies of the trigonometric examples are explicitly identified as finite illustrations, not substituted for the symbolic theorem.

The files `paper/figure_audit.md` and `paper/generated/figure-manifest.json` record data sources, hashes, and figure-specific claim status. The finite tables are generated from the same coordinate index used by verification. This makes a plot an inspectable derivative of evidence rather than an unsupported visual argument. The paper-specific build instructions accompany the LaTeX source.

Computer and AI assistance. The development and preparation of this manuscript used computer algebra, exact arithmetic, automated search, Lean, and AI assistance for research, code, and drafting. AI-generated assertions are not treated as mathematical evidence. The reported conclusions are tied to explicit proofs, exact witnesses, or separately scoped computational records. Alejandro Zarzuelo Urdiales is the sole named author of this manuscript; cited construction authors retain their mathematical attribution.

11 Consequences, limitations, and directions for further work

11.1 What the program has established

The strongest completed uniform construction in this project is Theorem 1.1, supplemented by the finite maximum in Proposition 5.1. Its significance is an explicit improvement over a standard all-order affine baseline together with a complete proof of real straight-line geometry, including the passage from local sign certificates to nonoverlapping cells. The original projective family and its mathematical antecedents remain attributed to Füredi and Palásti.

The earlier finite bounds survive independently of the incomplete March extension proof. The exterior construction now has a genuine geometric Lean theorem, and the failed repeated-exterior sequence explains why the proposed large gain cannot be obtained indefinitely by that method alone. The BBL work has progressed from counting and seed experiments to an actual one-step geometric theorem. Local fan geometry provides checked components for upper-bound arguments in the presence of multiple intersections.

The numerical refinement is small: it changes the constant term, while the benchmark deficit remains linear. Several published special families are stronger at their orders, and taking their maximum with any universal baseline produces another all-order function. Accordingly, neither “valid for every n ” nor “missing from an explicit table” is sufficient evidence of a state-of-the-art numerical claim. The most defensible completed contribution is a formal and explicit affine refinement, supported by a reusable geometric library and an attributed certificate catalog.

11.2 Complete the recursive geometric invariant

The next BBL theorem should return more than an existential lower bound. It should return sorted slopes, a nonduplicate triangle list, simplicity, saturation of the distinguished line, and

the central orientation condition. These are the data required to feed the construction into itself. Their preservation cannot be recovered merely from the number $T + q^2$ of triangles.

The exact integration tasks are identified: general support-permutation transport, conversion of the 21-line seed’s grouped labels to the tangent-grid order, retention of the central triangle independently of whether it appears in the chosen count list, and induction with a positive parameter threshold for each finite depth. A fixed positive perturbation parameter is not asserted to work at infinitely many depths. The stronger even companion additionally requires the quantitative visibility invariant. These tasks are geometric proof obligations, not missing numerical experiments.

11.3 Strengthen the nonsimple upper argument

The local multiplicity coefficients make triple and quadruple points the difficult cases. Large multiplicities already carry a favorable penalty in the proposed budget. The next formal step is a global extraction from an arbitrary arrangement into the checked local fans and a clean-line charging argument with all finite and unbounded incidences accounted for. A later improvement would need a sharper global interaction inequality for low multiplicities.

The exact local-resolution counterexamples rule out a blanket reduction to simple arrangements by triangle-preserving perturbation. A useful upper argument must therefore either retain singularities explicitly or prove a substantially more specific transformation theorem. The restricted consequences in the upper-bound section should guide that work without being advertised as a completed upper bound for the full classical problem.

11.4 Search beyond the tested neighborhoods

Large numbers of unsuccessful local proposals are useful when they identify which moves preserve the same obstruction. They are not an optimality certificate. The evidence favors testing singular configurations, different projective types, and more structured realization constraints instead of repeating the same smooth perturbations. Linear programming can help realize a fixed order type; oriented-matroid enumeration can organize combinatorial alternatives; exact arithmetic is then required to certify a proposed arrangement. The separate exactness boundaries of these tools should remain visible in subsequent work.

11.5 Resolve the remaining priority questions

Before claiming first numerical priority for G , a broader audit should inspect the relevant monograph chapters and follow-up literature on affine charts of the Füredi–Palásti arrangements. The original projective counts already explain why the constants are plausible. A complete formalization claim is supported by the checked development; a first-discovery claim requires a different kind of evidence.

For the finite cases, future comparisons should record not only explicit tables but also implicit consequences of published infinite constructions and elementary extensions. The author’s early attribution, a newly materialized exact witness, and the first appearance of a numerical inequality need not have the same date or source. Maintaining those distinctions allows the program to be continued without losing either its valid results or its research history.

Data and code availability. The LaTeX manuscript, bibliography, original vector figures, figure-generation scripts, proof-to-claim correspondence, exact coordinate catalog, negative experiments, and verification records accompany the public repository [29]. Publication of

this manuscript is not an assertion of external peer review. The proved results and the remaining conjectures are distinguished at their statements.

References

- [1] Nicolas Bartholdi, Jérémy Blanc, and Sébastien Loisel. On simple arrangements of lines and pseudo-lines in $\mathbb{P}^2$ and $\mathbb{R}^2$ with the maximum number of triangles. In *Surveys on Discrete and Computational Geometry*, volume 453 of *Contemporary Mathematics*, pages 105–116. American Mathematical Society, 2008. doi: 10.1090/conm/453/08797. URL <https://arxiv.org/abs/0706.0723>. Preprint first posted in 2007.
- [2] Jérémy Blanc. The best polynomial bounds for the number of triangles in a simple arrangement of n pseudo-lines. *Geombinatorics*, 21(1):5–14, 2011. URL <https://arxiv.org/abs/0801.2845>. Preprint first posted in 2008.
- [3] Gilles Clément and Johannes Bader. Tighter upper bound for the number of Kobon triangles, December 2007. URL <https://oeis.org/A006066/a006066.pdf>. Draft dated 21 December 2007.
- [4] Gábor Damásdi, Leonardo Martínez-Sandoval, Dániel T. Nagy, and Zoltán Lóránt Nagy. Triangle areas in line arrangements. *Discrete Mathematics*, 343:112105, 2020. doi: 10.1016/j.disc.2020.112105. URL <https://real.mtak.hu/114794/7/1-s2.0-S0012365X20302910-main.pdf>.
- [5] Leonardo de Moura and Sebastian Ullrich. The Lean 4 theorem prover and programming language. In *Automated Deduction—CADE 28*, Lecture Notes in Computer Science, pages 625–635. Springer, 2021. doi: 10.1007/978-3-030-79876-5_37.
- [6] Jeff Erickson. New lower bounds for convex hull problems in odd dimensions. *SIAM Journal on Computing*, 28(4):1198–1214, 1999. doi: 10.1137/S0097539797315410. URL <https://jeffe.cs.illinois.edu/pubs/pdf/convex.pdf>.
- [7] Stefan Felsner and Klaus Kriegel. Triangles in Euclidean arrangements. *Discrete & Computational Geometry*, 22(3):429–438, 1999. doi: 10.1007/PL00009471. URL <https://page.math.tu-berlin.de/~felsner/Paper/tri.pdf>.
- [8] David Forge and Jorge Luis Ramírez Alfonsín. Straight line arrangements in the real projective plane. *Discrete & Computational Geometry*, 20(2):155–161, 1998. doi: 10.1007/PL00009373.
- [9] Kobon Fujimura. *The Tokyo Puzzles*. Charles Scribner’s Sons, New York, 1978. ISBN 0684155362. Edited by Martin Gardner; translated by Fumie Adachi.
- [10] Zoltán Füredi and Ilona Palásti. Arrangements of lines with a large number of triangles. *Proceedings of the American Mathematical Society*, 92(4):561–566, 1984. doi: 10.1090/S0002-9939-1984-0760946-2. Also indexed by JSTOR DOI 10.2307/2045427.
- [11] Martin Gardner. *Wheels, Life, and Other Mathematical Amusements*. W. H. Freeman, New York, 1983. Kobon discussion cited in Clément–Bader at pp. 170–171 and 178.
- [12] Branko Grünbaum. *Arrangements and Spreads*, volume 10 of *CBMS Regional Conference Series in Mathematics*. American Mathematical Society, Providence, RI, 1972. doi: 10.1090/cbms/010.

- [13] HandWiki contributors. Kobon triangle problem, 2026. URL https://handwiki.org/wiki/Kobon_triangle_problem. Consulted 22 September 2026; a secondary record of the extension claim, not its proof.
- [14] Heiko Harborth. Some simple arrangements of pseudolines with a maximum number of triangles. *Annals of the New York Academy of Sciences*, 440(1):31–33, 1985. doi: 10.1111/j.1749-6632.1985.tb14536.x.
- [15] Ke Liang, Youming Liu, and Yurui Zhang. On the symmetries of optimal simple Kobon arrangements for $n = 28$. Zenodo preprint, version 4, July 2026. URL <https://zenodo.org/records/21181834>. Version-specific record, 3 July 2026; not an unrestricted exact-value theorem.
- [16] Andrea Maiorana. Exact fourteen-line Kobon configurations with 54 triangles. Public coordinate repository, 2026. URL https://github.com/rufio72/kobon_triangles_k14/tree/e47c7cfd9661e54d29befb83118bf83d5f804028. Descriptive title; commit e47c7cfd9661e54d29befb83118bf83d5f804028, 12 August 2026; CC BY 4.0.
- [17] OEIS Foundation Inc. Entry A006066: Kobon triangles. The On-Line Encyclopedia of Integer Sequences, 2026. URL <https://oeis.org/A006066>. Consulted 22 September 2026; revision 245 and its attribution comments audited.
- [18] Roman Parpalak and Denis Utkin. The $18 \cdot 2^t + 1$ triangle-maximal series of straight lines, 2026. URL <https://arxiv.org/abs/2604.22035>. Version 1, 23 April 2026.
- [19] Roman Parpalak and Denis Utkin. The 8–14–26 even series. Public research draft and construction repository, 2026. URL <https://github.com/parpalak/kobon-even-draft/blob/4da100acd30d9fb2c5b2a1fa3fb9ce848a2e52a5/8-14-26-even-series.md>. Snapshot at commit 4da100acd30d9fb2c5b2a1fa3fb9ce848a2e52a5; descriptive repository title.
- [20] Roman Parpalak and Denis Utkin. Kobon solutions: coordinate gallery. Public data repository, 2026. URL <https://github.com/ud1/kobon-solutions/tree/feae4f5571a988a3ed45bd26a36259ebb7d7d3d2>. Snapshot at commit feae4f5571a988a3ed45bd26a36259ebb7d7d3d2, 21 September 2026; original individual credits retained.
- [21] ProofAtlas. Kobon triangle problem: collaboration research page, 2026. URL <https://www.proofatlas.ai/collaboration/kobon-triangle-problem/>. Consulted 22 September 2026; reported restricted results were not independently replayed.
- [22] G. B. Purdy. Triangles in arrangements of lines. *Discrete Mathematics*, 25(2):157–163, 1979. doi: 10.1016/0012-365X(79)90018-9.
- [23] George Purdy. Triangles in arrangements of lines. II. *Proceedings of the American Mathematical Society*, 79(1):77–81, 1980. doi: 10.1090/S0002-9939-1980-0560588-4.
- [24] Jean-Pierre Roudneff. The maximum number of triangles in arrangements of pseudolines. *Journal of Combinatorial Theory, Series B*, 66(1):44–74, 1996. doi: 10.1006/jctb.1996.0006.
- [25] Pavlo Savchuk. Constructing optimal Kobon triangle arrangements via table encoding, SAT solving, and heuristic straightening, 2025. URL <https://arxiv.org/abs/2507.07951>. Version 1 inspected.

- [26] Thomas O. Strommer. Triangles in arrangements of lines. *Journal of Combinatorial Theory, Series A*, 23(3):314–320, 1977. doi: 10.1016/0097-3165(77)90022-X.
- [27] The mathlib Community. The Lean mathematical library. In *Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs*, pages 367–381. ACM, 2020. doi: 10.1145/3372885.3373824. URL <https://leanprover-community.github.io/papers/mathlib-paper.pdf>.
- [28] Alejandro Zarzuelo Urdiales. New lower bounds for even Kobon numbers. Archivara preprint, March 2026. URL <https://archivara.org/paper/48b411c9-0e03-4592-931e-179b9a1c2312>. Original March result; the archived general extension proof was incomplete.
- [29] Alejandro Zarzuelo Urdiales. Verified Kobon constructions: research and Lean proof repository. Source code, exact certificates, and research records, 2026. URL <https://github.com/alejandrozu/kobon-proof>. Research state reviewed through 22 September 2026.

A Complete saved finite table, orders 3–60

The entries below are the strongest saved coordinate lower bounds in the frozen release. They are separated by arrangement class. The classical column may use multiple intersections; the simple column may not. The OEIS column is a dated explicit-table comparison, not an exhaustive priority audit. The upper columns reproduce attributed external benchmarks and are not formal upper-bound conclusions of this project. A difference between a lower and upper entry represents an unresolved gap within the stated scope, not a proof that intermediate values are unattainable.

Table 10: Saved finite lower bounds, separated by arrangement class, and attributed comparison upper values. OEIS is the explicit table snapshot of 20 September 2026; a dash means omitted. The upper columns are external results, not upper theorems proved by this project.

n	$G(n)$	Saved simple	Saved classical	OEIS lower	Blanc simple	U_T
3	1	1	1	1	1	1
4	2	2	2	2	2	2
5	5	5	5	5	5	5
6	7	7	7	7	7	8
7	11	11	11	11	11	11
8	14	14	15	15	14	16
9	20	21	21	21	21	21
10	24	25	25	25	25	26
11	31	32	32	32	33	33
12	37	37	38	38	38	40
13	45	47	47	47	47	47
14	52	53	54	54	53	56
15	62	65	65	65	65	65
16	70	72	72	72	72	74
17	81	85	85	85	85	85
18	91	93	93	93	93	96
19	103	107	107	107	107	107
20	114	116	117	117	116	120
21	128	133	133	133	133	133
22	140	143	143	143	143	146
23	155	161	161	161	161	161
24	169	172	172	172	172	176
25	185	191	191	191	191	191
26	200	203	204	204	203	208
27	218	225	225	225	225	225
28	234	238	238	238	238	242
29	253	261	261	261	261	261
30	271	275	275	275	275	280
31	291	299	299	299	299	299
32	310	314	315	315	314	320
33	332	341	341	341	341	341
34	352	357	357	357	357	362
35	375	385	385	385	385	385
36	397	402	402	402	402	408
37	421	431	431	431	431	431
38	444	449	450	450	449	456
39	470	470	470	–	481	481
40	494	494	494	–	500	506
41	521	533	533	533	533	533
42	547	553	553	553	553	560

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n	$G(n)$	Saved simple	Saved classical	OEIS lower	Blanc simple	U_T
43	575	587	587	587	587	587
44	602	608	608	—	608	616
45	632	645	645	645	645	645
46	660	667	667	667	667	674
47	691	691	691	—	705	705
48	721	721	721	—	728	736
49	753	767	767	767	767	767
50	784	791	792	792	791	800
51	818	818	818	—	833	833
52	850	850	850	—	858	866
53	885	885	885	—	901	901
54	919	919	919	—	927	936
55	955	955	955	—	971	971
56	990	990	990	—	998	1008
57	1028	1045	1045	—	1045	1045
58	1064	1073	1073	—	1073	1082
59	1103	1103	1103	—	1121	1121
60	1141	1141	1141	—	1150	1160

B The complete 138-certificate catalog

Each row identifies a finite arrangement and its certified triangle count. A hash distinguishes coordinate identities even when n and T coincide. The complete coordinates, support triples, source paths, original metadata, and Lean theorem names are machine-readable in the accompanying catalog. Printing hundreds of large rational coefficients would obscure rather than improve the mathematical presentation; the hash-indexed files provide their exact reproducible specification.

The catalog contains 138 distinct promoted coordinate identities, not 138 new numerical records. S denotes a simple arrangement; M denotes a verified arrangement with multiple intersections. Every row has a classical lower-bound theorem; S rows also have simplicity certificates. The hash is a prefix of the canonical coordinate SHA-256 identity, not of the source file. Full source paths, source-file hashes, theorem modules, and complete identities are supplied in `paper/generated/certificate-catalog.json`.

Source codes identify repository collections, not discovery authors: FT, retained finite table (mixed published and derived sources); EX, exact extension; HI, historical work inventory; HY, compatible-seed/hybrid work; SC, current session (phase/chart or classical dyadic construction, distinguished in the JSON metadata); MA, Andrea Maiorana’s external certificates; PU, Parpalak–Utkin gallery inputs; KR, earlier research collection; OT, other documented source. Published constructions retain their original attribution. Native-evaluation trust boundaries are discussed in the text.

Table 11: Complete promoted coordinate catalog. Repeated orders and counts can have distinct coordinate witnesses; no numerical priority follows from inclusion.

ID	n	T	Class	Collections	Coordinate identity
1	3	1	S	FT	05c82de62571
2	4	2	S	FT	0e541bd5bbe7
3	5	5	S	FT	948a0bdf2dd3
4	6	7	S	FT,EX,HI	6ea9ebaa7bf2

Continued from previous page

ID	n	T	Class	Collections	Coordinate identity
5	7	11	S	FT	e6b9a626a163
6	8	14	S	FT	2a63f4d9308d
7	9	21	S	FT	6ebbed82d063
8	10	25	S	FT	b767c427fa30
9	11	32	S	FT, HY	b955e7c9923a
10	12	38	M	FT	38f69f380097
11	13	47	S	FT	ddec1f788550
12	14	53	S	FT	45157b2f20fe
13	15	65	S	FT	4d3bf3b46db5
14	16	72	S	FT	391cf79afe24
15	17	85	S	FT	7cf3c061a447
16	18	93	S	FT	edb6246a6de7
17	19	107	S	FT	a2f32bdbcd1e
18	20	116	S	FT, EX, HI	adc03eed024d
19	21	133	S	FT	36bae754f198
20	22	143	S	FT	54396c646a7b
21	23	161	S	FT	969ad7d3adaf
22	24	172	S	FT	e63e906b9903
23	25	191	S	FT	b568bffb5a19
24	26	203	S	FT	e61b423c7c3d
25	27	225	S	FT	7d6ae9a04c46
26	28	238	S	FT	56ef2acb9edd
27	29	261	S	FT	d4805c25a121
28	30	275	S	FT	dcfe649a79cf
29	31	299	S	FT	70f23b9463cb
30	32	314	S	FT	9ab78b00c233
31	33	341	S	FT	95f3b770119e
32	34	357	S	FT	fbde0a760c6e
33	35	385	S	FT	9ee99af6b980
34	36	402	S	FT	3fc01b7d2a1c
35	37	431	S	FT	11f0bde7bac3
36	38	449	S	FT	46ad5a32af83
37	39	469	M	FT, HI	cf24d7d33c00
38	40	494	S	FT	785374ca691b
39	41	533	S	FT	01e2ef66f8fd
40	42	553	S	FT	2bdc9d67365a
41	43	587	S	FT	861d8be27dee
42	44	608	S	FT, HI	b5277675e356
43	45	645	S	FT	3dc6e882e9b8
44	46	667	S	FT, PU	32b8ca2f231d
45	47	690	S	FT	d8af27fc13d1
46	48	720	S	FT	5e9767e73b31
47	49	767	S	FT	957e6f151a57
48	50	791	S	FT	2ea22b558bf8
49	51	817	M	FT, HI, KR	bd215f67ab10
50	52	850	S	FT	4b0d823a1fa8
51	53	884	S	FT	723d7eff8d45
52	54	918	S	FT	915abb8ed17c
53	55	954	S	FT	ea7675d9b8a3
54	56	990	S	FT	de8a415c9b29
55	57	1045	S	FT	9a1d337c75ba
56	58	1073	S	FT	d988db552530
57	59	1102	S	FT	fd3b4280376e
58	60	1140	S	FT	16af6d0b260e

Continued from previous page

ID	n	T	Class	Collections	Coordinate identity
59	12	37	S	FT, HY	8c718a334b7b
60	39	468	S	FT	5c5b0b26ee22
61	51	816	S	FT	6fa6b7d91d28
62	5	3	S	EX	6265d6a7c557
63	6	4	S	EX	7e060e9a0ac1
64	19	107	S	EX	5d2a7588c1e2
65	21	126	S	EX	f18756b72791
66	5	5	S	EX, HI	dd0748b7cdfb
67	7	10	S	EX, HI	4b5bb6e99d62
68	21	132	S	HY	3d8bae2fa729
69	22	142	S	HY	e683945e66cd
70	41	532	S	HY	6d48a3411f41
71	42	552	S	HY	7b29a6bd1191
72	81	2132	S	HY	7bc6b303744f
73	82	2172	S	HY	d75c9101c8a5
74	161	8532	S	HY	230691d59eee
75	162	8612	S	HY	1512e6966074
76	9	19	M	HI	e16410b887ee
77	15	61	M	HI	7c38d54f9e52
78	21	127	M	HI	b8dcff224cdb
79	27	217	M	HI	6670a2b14284
80	33	331	M	HI	25fd8f0c3311
81	48	715	M	HI	ba6f5baae76d
82	99	3169	M	HI, KR	4fef38d87d0c
83	195	12481	M	HI, KR	cd3497dbf846
84	195	12480	S	HI	1b9230fb738a
85	44	602	S	HI	0b0250bf6213
86	99	3168	S	HI	a0ea30825a73
87	39	470	S	SC	30caa6f7f894
88	47	691	S	SC	fac44af18c55
89	48	721	S	SC	2fb5ef0df4cf
90	51	818	S	SC	c18e1baf25f5
91	53	885	S	SC	6bc5b14d0b68
92	54	919	S	SC	a254c1b4f05b
93	55	955	S	SC	202625a39085
94	59	1103	S	SC	ba482375103b
95	60	1141	S	SC	9bc5096cc3fd
96	65	1365	S	SC	b1dcbeb4ff39
97	66	1397	S	SC	327ecbb9b7ba
98	99	3170	S	SC	ea507e2de137
99	129	5461	S	SC	6dae0012aa49
100	130	5525	S	SC	e0559b54aee5
101	195	12482	S	SC	5413dd7b60ee
102	14	54	M	MA	d47aea63afc1
103	14	54	M	MA	18bbcff4f433
104	14	54	M	MA	78f750c071a5
105	14	54	M	MA	912da8ec9bc7
106	14	54	M	MA	9134999072a9
107	14	54	M	MA	9c871e103a46
108	14	54	M	MA	b3dcae0bc1d8
109	14	54	M	MA	9219cdaa6f90
110	14	54	M	MA	d2babc1b9431
111	14	54	M	MA	2b34ca22dc89
112	14	54	M	MA	fa427b3d27ef

Continued from previous page

ID	n	T	Class	Collections	Coordinate identity
113	14	54	M	MA	1ec1fd5cb2ee
114	14	54	M	MA	3e5d182f8408
115	14	54	M	MA	17f8c0524eb1
116	14	54	M	MA	c472f97237f1
117	14	52	S	MA	2eb83dc242a1
118	14	53	S	MA	5e6068f5003e
119	14	52	S	MA	b3f4749f6dee
120	14	52	S	MA	c6615430ae3d
121	8	14	S	PU	ebff58687458
122	14	51	S	PU	d67e7a6b3b08
123	20	114	S	PU	bc312fc85ea5
124	26	203	S	PU	5452b4ffa1a9
125	32	313	S	PU	018f9fc6b73f
126	38	448	S	PU	7d59f504aba8
127	50	791	S	PU	6bde86a3dcdb
128	8	15	M	PU	a138081033c3
129	14	54	M	PU	105765145381
130	20	117	M	PU	382171ac62cc
131	26	204	M	PU	f71379c4db03
132	32	315	M	PU	5914f2bb88d1
133	36	402	S	PU	ce43a1077597
134	38	450	M	PU	8ba2bdb14e85
135	42	553	S	PU	9849e255d5da
136	50	792	M	PU	3d40cb582616
137	19	107	S	OT	6e8ca6f56a85
138	6	6	M	OT	af8f599eb8a1

C Historical inventory of the author's outputs

This table preserves the earlier research inventory, including bounds later superseded by G , by a stronger retained certificate, or by known literature constructions. Its original labels describe how a number entered that inventory. They do not retroactively establish a missing general geometric argument. The active result aliases independently verify the retained numerical inequalities. Published inputs and older matching results keep their original attribution.

This is a historical numerical inventory, not a list of original discoveries. B means an elementary base case; C means an explicit coordinate construction in that inventory; E means the then-evaluated defect-extension consequence (called G in the historical file, renamed here to avoid confusion with the present function G). The numerical inequalities are now retained independently by the compiled `Results.earlier_NNN` aliases. This does not formalize all premises of the former general defect argument or establish priority. The right columns display current, separately proved bounds.

Table 12: Earlier work inventory compared with the present unconditional formula and saved certificate envelope.

n	Earlier output	Historical basis	Current $G(n)$	Best saved classical
3	1	B	1	1
4	2	E	2	2
5	5	B/C	5	5
6	7	C	7	7

Continued from previous page

n	Earlier output	Historical basis	Current $G(n)$	Best saved classical
7	10	C	11	11
8	14	E	14	15
9	19	C	20	21
10	25	E	24	25
11	28	E	31	32
12	36	E	37	38
13	39	E	45	47
14	53	E	52	54
15	61	C	62	65
16	72	E	70	72
17	76	E	81	85
18	93	E	91	93
19	98	E	103	107
20	116	C	114	117
21	127	C	128	133
22	143	E	140	143
23	149	E	155	161
24	172	E	169	172
25	178	E	185	191
26	203	E	200	204
27	217	C	218	225
28	238	E	234	238
29	245	E	253	261
30	275	E	271	275
31	283	E	291	299
32	314	E	310	315
33	331	C	332	341
34	357	E	352	357
35	366	E	375	385
36	402	E	397	402
37	411	E	421	431
38	449	E	444	450
39	469	C	470	470
40	470	E	494	494
41	496	E	521	533
42	553	E	547	553
43	564	E	575	587
44	608	C	602	608
45	618	E	632	645
46	667	E	660	667
47	679	E	691	691
48	715	C	721	721
49	722	E	753	767
50	791	E	784	792
51	817	C	818	818
52	818	E	850	850
53	852	E	885	885
54	886	E	919	919
55	920	E	955	955
56	956	E	990	990
57	992	E	1028	1045
58	1073	E	1064	1073
59	1088	E	1103	1103
60	1104	E	1141	1141

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n	Earlier output	Historical basis	Current $G(n)$	Best saved classical
99	3169	C	3170	3170
195	12481	C	12482	12482

D Evidence navigation and excluded drafts

The following stable repository paths locate the complete evidence. This appendix is a navigation aid, not an additional source of axioms.

Path or collection	Contents and role
<code>archive/2026-03-original/</code>	Original March sources, preserved as historical evidence; incomplete and excluded from the active build.
<code>Kobon/</code>	Active Lean geometry, counts, transformations, certificates and formal proof audit.
<code>research/kobon-extension/</code>	Both-parity extension manuscript, boundary-defect analysis and examples.
<code>research/successor-formalization/</code>	Recurrence status, 49-seed numerical target, and exterior-resource obstruction.
<code>research/kobon-hybrid/</code>	Compatible eleven-line seed, interval argument and sparse family manuscript.
<code>research/kobon-research/</code> <code>research/kobon-own-results/</code>	Additional-line family draft and finite witnesses. Historical inventory, candidate comparisons and priority audit.
<code>research/all-n-formalization/</code>	Earlier and current all-order formulas, finite enhancement table and interpretation.
<code>research/six-hour-2026-09-21/</code>	Three-hour research release, source audits, local upper arguments, family components and cumulative review. The directory keeps its original allocation name.
<code>experiments/2026-09-20/</code> and <code>experiments/2026-09-21/</code> <code>verification/</code>	Frozen search scripts, parameters, exact outputs, failed approaches and interrupted-run status. Build logs, theorem audit, coordinate recounts, source hashes and the interval summary.
<code>evidence/sources.json</code>	Bibliographic roles, versions, immutable commits and provenance.
<code>paper/</code>	This manuscript, bibliography, original figures, generated tables and reproduction instructions.

The uncompiled real-parameter 33-line draft is separately labelled. The large 321-, 322-, and 641-line exact computational witnesses are not included among the 138 Lean-promoted coordinate identities. The interrupted direct recount at 642 lines remains incomplete evidence. None is used as an assumption in Theorem 1.1 or Proposition 5.1.